



CSA/ANSI LC 1:23 • CSA 6.26:23
National Standard of Canada
American National Standard



Fuel gas piping systems using corrugated stainless steel tubing



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Preface

This is the seventh edition of CSA/ANSI LC 1 • CSA 6.26, *Fuel gas piping systems using corrugated stainless steel tubing*. It supersedes the previous editions published in 2019, 2018, 2016, 2014, 2005, and 1997.

This edition of this Standard has been updated to reflect current industry practice, and address CSA Group's guide on drafting Standards. Major changes to this edition include the following:

- a) this Standard has been revised for accessibility;
- b) installation instructions using manifolds have been revised;
- c) in Clause [5.16.4](#), the test specimen sizes have been updated; and
- d) Annex [C](#) has been updated to bring the recommendations of this non-mandatory annex into agreement with the bonding requirements on the current editions of the US and Canadian model codes.

This Standard is considered suitable for use for conformity assessment within the stated scope of the Standard.

This Standard was prepared by the Joint Subcommittee on Corrugated Stainless Steel Tubing, under the jurisdiction of the Technical Committee on Gas Appliances and Related Accessories, the Z21/83 Technical Committee on Performance and Installation of Gas Burning Appliances and Related Accessories, and the Strategic Steering Committee on Fuels and Appliances, and has been formally approved by the Technical Committees and the Interprovincial/Territorial Gas Advisory Council.

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- 1) *Use of the singular does not exclude the plural (and vice versa) when the sense allows.*
- 2) *Although the intended primary application of this Standard is stated in its Scope, it is important to note that it remains the responsibility of the users of the Standard to judge its suitability for their particular purpose.*
- 3) *This Standard was developed by consensus, which is defined by CSA Policy governing standardization — Code of good practice for standardization as "substantial agreement. Consensus implies much more than a simple majority, but not necessarily unanimity". It is consistent with this definition that a member may be included in the Technical Committee list and yet not be in full agreement with all clauses of this Standard.*
- 4) *To submit a request for interpretation of this Standard, please send the following information to inquiries@csagroup.org and include "Request for interpretation" in the subject line:*
 - a) *define the problem, making reference to the specific clause, and, where appropriate, include an illustrative sketch;*
 - b) *provide an explanation of circumstances surrounding the actual field condition; and*
 - c) *where possible, phrase the request in such a way that a specific "yes" or "no" answer will address the issue.*

Committee interpretations are processed in accordance with the CSA Directives and guidelines governing standardization and are available on the Current Standards Activities page at standardsactivities.csa.ca.

- 5) *This Standard is subject to review within five years from the date of publication. Suggestions for its improvement will be referred to the appropriate committee. To submit a proposal for change, please send the following information to inquiries@csagroup.org and include "Proposal for change" in the subject line:*
- a) *Standard designation (number);*
 - b) *relevant clause, table, and/or figure number;*
 - c) *wording of the proposed change; and*
 - d) *rationale for the change.*

CSA/ANSI LC 1:23 • CSA 6.26:23

Fuel gas piping systems using corrugated stainless steel tubing

0 History of CSA/ANSI LC 1 • CSA 6.26

In 1983, the Gas Research Institute (GRI) initiated its “Residential/Commercial Piping Program”. This research and development effort was aimed at identifying and developing innovative building piping systems and materials as a viable alternative to conventional rigid black iron piping.

One of the most promising concepts that emerged from the GRI project was a piping system using semi-flexible, corrugated stainless steel tubing in conjunction with elevated gas pressures [up to 5 psig (34.5 kPa)]. This system offered several advantages over rigid black iron piping systems, including ease and speed of installation, elimination of the need for precise on-site measuring, cutting and threading of piping sections, and elimination of the need for certain fittings such as elbows, tees, and couplings. A disadvantage was increased flow resistance imposed by the tubing corrugations and smaller internal diameters. This resulted in the requirement for higher system pressures and an additional gas pressure regulator upstream of equipment requiring lower supply pressures.

In November 1986, Foster-Miller Inc., a GRI contractor assigned to the piping system project, requested the American Gas Association (A.G.A.) Laboratories to develop construction and performance criteria for use as the basis for a third-party certification program for corrugated stainless steel piping systems. This work was undertaken and, on September 8, 1987, the “A.G.A. Requirements For Natural Gas Piping Systems Using Corrugated Stainless Steel Conduit”, No. 1-87 was published.

A.G.A. 1-87 was developed with input from Foster-Miller Inc. who had conducted numerous testing programs and gained considerable experience with these types of piping systems as part of the GRI project. Information was also drawn from existing standards which cover similar types of products. Safety issues, such as potential damage to the tubing caused by bending, stretching, torquing, crushing, or impacting, and accidental puncturing of concealed tubing, were given prime consideration in developing the construction and performance criteria contained in this document. Emphasis also was placed on comprehensive instructions to guide the installer through proper step-by-step installation and checkout procedures.

A.G.A. 1-87 was referenced in the fourth edition of the *National Fuel Gas Code*, ANSI Z223.1, as a recognized document for testing and listing corrugated stainless steel gas piping systems.

As interest in using these piping systems increased, there was some reluctance by local code authorities to accept systems not covered by a nationally recognized safety standard. Consequently, the A.G.A. Laboratories, in August 1989, applied to the American National Standards Institute (ANSI) for recognition as an approved sponsor under the canvass method for developing ANSI standards. On December 8, 1989, the A.G.A. Laboratories was approved by ANSI as a canvass sponsor. Consequently, A.G.A. 1-87 was retitled as proposed *American National Standard for Fuel Gas Piping Systems Using Corrugated Stainless Steel Tubing*, AGA LC 1, and was distributed to all known interested parties for comment in accordance with the ANSI procedures for standards development under the canvass method.

The first edition of this Standard was approved by the American National Standards Institute, Inc., on January 25, 1991.

With the onset of the Free Trade Agreement between the United States and Canada on January 2, 1988, significant attention was given to the harmonization of the United States and Canadian safety standards addressing gas-fired equipment for residential, commercial, and industrial applications. It was believed that the elimination of the differences between the standards would remove potential trade barriers and provide an atmosphere in which North American manufacturers could market more freely in the United States and Canada. The harmonization of these standards was also seen as a step toward harmonization with international standards. Joint subcommittees were established to facilitate the standards harmonization process between the United States and Canada.

The harmonized draft standard was based on coverage from the first edition of ANSI/AGA LC 1-1991, and Addenda, LC 1a-1993, and LC 1b-1994.

The harmonized draft standard was processed under the ANSI canvass method, with a final ballot on its acceptance as a proposed ANSI standard dated December 9, 1994.

The proposed first edition of ANSI/IAS LC 1 • CGA 6.26 standard was approved by the CGA Standards Advisory Committee on November 3, 1997, by the Interprovincial/Territorial Gas Advisory Council on September 18, 1997, and by the American National Standards Institute on October 28, 1996.

The first edition of the harmonized gas piping systems using corrugated stainless steel tubing standard was approved by the Standards Advisory Committee and the Standards Council of Canada on September 18, 1997, and by the American National Standards Institute, Inc., on October 28, 1996.

The second edition of the harmonized gas piping systems using corrugated stainless steel tubing standard was approved by the Standards Advisory Committee and the Standards Council of Canada on October 27, 2004, and by the American National Standards Institute, Inc., on March 9, 2005.

The third edition of the harmonized gas piping systems using corrugated stainless steel tubing standard was approved by the Interprovincial/Territorial Gas Advisory Council on March 20, 2014, and by the American National Standards Institute, Inc. on September 17, 2013.

The fourth edition of the harmonized gas piping systems using corrugated stainless steel tubing standard was approved by the Interprovincial/Territorial Gas Advisory Council on March 27, 2016, and by the American National Standards Institute, Inc. on March 10, 2016.

The fifth edition of the harmonized gas piping systems using corrugated stainless steel tubing standard was approved by the Interprovincial/Territorial Gas Advisory Council on January 9, 2018, and by the American National Standards Institute, Inc. on January 24, 2018.

This, the sixth edition of the harmonized gas piping systems using corrugated stainless steel tubing standard, was approved by the Interprovincial/Territorial Gas Advisory Council on September 26, 2019, and by the American National Standards Institute, Inc. on November 11, 2019.

This, the seventh edition of the harmonized gas piping systems using corrugated stainless steel tubing standard, was approved by the Interprovincial/Territorial Gas Advisory Council on mm/dd/year, and by the American National Standards Institute, Inc. on mm/dd/year.

ANSI/IAS LC 1-1997 • CGA 6.26-M97 ANSI LC 1-2005 • CSA 6.26-2005
 ANSI/IAS LC 1a-1999 • CGA 6.26a-M99 ANSI LC 1a-2009 • CSA 6.26a-2009
 ANSI/IAS LC 1b-2001 • CGA 6.26b-2001 ANSI LC 1b-2011 • CSA 6.26b-2011
 ANSI LC 1-2014 • CSA 6.26-2014 ANSI LC 1-2016 • CSA 6.26-2016
 ANSI LC 1-2018 • CSA 6.26-2018 CSA/ANSI LC 1:19 CSA 6.26:19

The following identifies the designation and year of this edition of the standard:
 CSA/ANSI LC 1:22 • CSA 6.26:22

1 Scope

1.1 Inclusions

This Standard applies to fuel gas piping systems using corrugated stainless steel tubing (CSST), intended for installation in residential, commercial, or industrial buildings, and including the following components as a minimum:

- a) corrugated stainless steel tubing (CSST) piping systems not exceeding a size of 2 in (50.8 mm), based on the nominal inside diameter of the tubing;
- b) fittings for connection to the CSST; and
- c) striker plates (see Clause 3) to protect the installed CSST from puncture threats.

Other components of piping systems covered by this Standard include gas manifolds, gas pressure regulators, manual gas valves, quick-disconnect devices, excess flow valves, and gas convenience outlets (see Clauses 4.3, 4.4, and 4.5).

Note: *If such additional components are required to complete the piping system installation, they are either provided as part of the piping system or specified in the manufacturer's installation instructions [see Clause 4.7 n)].*

1.2 Additional inclusions

This Standard also applies to the following:

- a) CSST piping systems in which portions of the piping are exposed to the outdoors, as required, to make connections to outdoor gas meters or to outdoor gas appliances, which are attached to, mounted on, or located in close proximity to the building structure; and
- b) CSST piping systems that are used in conjunction with other approved fuel gas piping materials.

1.3 Exclusions

This Standard does not apply to the following:

- a) CSST, whether coated or uncoated, intended for direct burial underground. Tubing is installed underground only when encased inside either an approved conduit or as part of an approved engineered system and in accordance with local codes and the manufacturer's instructions; and
- b) gas connectors for appliances. These connectors are covered by CSA/ANSI Z21.24 • CSA 6.10 and ANSI Z21.69 • CSA 6.16.

1.4 Concealment of tube fittings

This Standard includes criteria to establish the suitability of concealed mechanical tube fittings for use with concealed gas piping (see Clause 3).

1.5 System pressure ratings

This Standard applies to piping systems rated at

- a) 5 psi (34.5 kPa) and intended for exposure to maximum actual operating pressures (see Clause 3) not exceeding 6.5 psi (44.8 kPa); or
- b) 25 psi (172.5 kPa) and intended for exposure to maximum actual operating pressures not exceeding 30 psi (207 kPa).

Note: *The manufacturer has the option to certify its system to either pressure rating.*

1.6 Applicable fuels

This Standard applies to piping systems intended for use only with fuel gas as defined in Clause 3, but does not apply to piping systems intended for use with propane in the liquid state.

1.7 Pressure references

All references to psi throughout this Standard are to be considered gauge pressure unless otherwise specified.

1.8 Measurement units

If a value for measurement as given in this Standard is followed by an equivalent value in other units, the first stated value is to be regarded as the specification.

Note: *This Standard contains SI (metric) units corresponding to the yard/pound quantities, the purpose being to allow this Standard to be used in SI (metric) units. IEEE/ASTM SI 10 or ISO 80000-1 is used as a guide in making metric conversion from yard/pound quantities. If a value for a measurement and a corresponding value in other units are both specified as a quoted marking requirement, the first stated unit, or both, are to be provided. Given corresponding values might be approximate. See also Annex D.*

1.9 Terminology

In this Standard, “shall” is used to express a requirement, i.e., a provision that the user shall satisfy in order to comply with the Standard; “should” is used to express a recommendation or that which is advised but not required; and “may” is used to express an option or that which is permissible within the limits of the Standard.

Notes accompanying clauses do not include requirements or alternative requirements; the purpose of a note accompanying a clause is to separate from the text explanatory or informative material.

Notes to tables and figures are considered part of the table or figure and may be written as requirements.

Annexes are designated normative (mandatory) or informative (non-mandatory) to define their application.

2 Reference publications

This Standard refers to the following publications. For dated references, only the edition cited shall apply. For undated references, the latest edition of the referenced document (including any amendments) shall apply.

CSA Group

ANSI Z21.24-2015 • CSA 6.10-2015

Connectors for gas appliances

ANSI Z21.41-2014 • CSA 6.9-2014

Quick disconnect devices for use with gas fuel appliances

ANSI Z21.69-2015 • CSA 6.16-2015

Connectors for moveable gas appliances

ANSI Z21.90-2015 • CSA 6.24-2015

Gas convenience outlets and optional enclosures

B149.1

Natural gas and propane installation code

C22.1

Canadian Electrical Code, Part I

ASTM International

B117-18

Standard Practice for Operating Salt Spray (Fog) Apparatus

D746-14

Standard Test Method for Brittleness Temperature of Plastics and Elastomers by Impact

E84-19a

Standard Test Method for Surface Burning Characteristics of Building Materials

ASME (The American Society of Mechanical Engineers)

B1.20.1-2013 (R2018)

Pipe Threads, General Purpose (Inch)

IAPMO (International Association of Plumbing and Mechanical Officials)

IAPMO/ANSI UPC 1-2018

Uniform Plumbing Code

ICC (International Code Council)

ICC/ANSI 2.0

Manufactured Housing Construction and Safety Standards (MHCSS)

IFGC

International Fuel Gas Code

IEEE (Institute of Electrical and Electronics Engineers)/ASTM International

IEEE/ASTM SI 10-2016

American National Standard for Metric Practice

ISO (International Organization for Standardization)

80000-1:2009

Quantities and units — Part 1: General

NFPA (National Fire Protection Association)

58

Liquefied Petroleum Gas Code

70

National Electrical Code

501A

*Standard for Fire Safety Criteria for Manufactured Home Installations, Sites, and Communities***NFPA (National Fire Protection Association)/AGA (American Gas Association)**

NFPA 54-2018/ANSI Z223.1-2018

*National Fuel Gas Code***ULC (Underwriters Laboratories of Canada)**

CAN/ULC-S102:2018

*Standard Method of Test for Surface Burning Characteristics of Building Materials and Assemblies***United States Code of Federal Regulations**

24 CFR Part 3280:2021

Federal Manufactured Home Construction and Safety Standard

3 Definitions

The following definitions shall apply in this Standard:

Appliance (equipment) — any device that uses gas as a fuel or raw material to produce light, heat, power, refrigeration, or air conditioning.

Appliance connector — a factory-fabricated assembly of gas conduit and related fittings designed to convey gaseous fuel, and used for making connections between a gas supply piping outlet and the gas inlet to an appliance. It is equipped at each end for attachment to standard taper pipe threads.

Approved — acceptable to the AHJ.

Authority having jurisdiction (AHJ) — the organization, office, or individual responsible for approving equipment, an installation, or a procedure.

Coating — dipping or painting of the tube with a material that is not removable without damage to the tubing.

Concealed gas piping — gas piping that, when in place in a finished building, would require removal of permanent construction to gain access to the piping.

Equivalent hydraulic diameter (EHD) — a theoretical size that reflects the hydraulic performance of the tubing. It is not a true physical measurement.

Flow capacity — as used in this Standard, the amount of a specified gas corrected to standard conditions of 30 in WC and 60 °F (760 mm Hg dry and 15.6 °C) that will flow through a specified length and configuration of tubing, a manifold, fitting, or other component at a specified pressure drop in a fixed period of time.

Fuel gas — a commercially distributed gas used for combustion including natural gas, propane gas-air mixtures, or mixtures of these gases.

Gas utilization equipment — any device that uses gas as a fuel or raw material, or both.

Jacket/covering — a sheath that is fabricated over the tubing without adhering to the surface or filling into all the convolutions and is easily removed without damage to the surface of the tubing.

Listed — equipment or materials included in a list published by an organization

- a) acceptable to the AHJ;
- b) concerned with product evaluation that maintains periodic inspection of production of listed equipment or materials; and
- c) whose listing states either that the equipment or material meets appropriate standards or has been tested and found suitable for use in a specified manner.

Maximum actual operating pressure — the maximum pressure existing in a piping system, including transients, during a normal annual operating cycle. The maximum actual operating pressure normally occurs at regulator lockup under no-flow conditions.

Piping system — as used in this Standard, an assembly of corrugated stainless steel tubing (CSST), tubing connection fittings, shutoff valve(s), tube shielding devices, and other listed components, intended for field assembly and installation, to distribute fuel gas to residential and commercial gas utilization equipment.

Pressure drop — the loss in static pressure of flowing fuel gas due to friction or other flow resistance in tubing, fittings, valves, regulators, or other devices in the piping system.

Pressure regulator — a device placed in a gas line for reducing, controlling, and maintaining the pressure in that portion of the piping system downstream of the device.

Qualified installer — any individual, firm, corporation, or company that, either in person or through a representative,

- a) is engaged in and responsible for the installation or replacement of building gas piping systems;
- b) is experienced in such work and familiar with all precautions required; and
- c) has complied with all the requirements of the AHJ.

Quick-disconnect device — a hand-operated device that provides a means for connecting and disconnecting an appliance or an appliance connector to a gas supply, and that is equipped with an automatic means to shut off the gas supply when the device is disconnected.

Shielding device — a component of the piping system used to protect the installed corrugated tubing from accidental puncture by nails, screws, or similar hardware at concealed tubing support points.

Shutoff valve — a device used in piping to control the gas supply to any section of the piping system or to an appliance.

Striker plate — a special type of shielding device used when concealed tubing is run through wall studs, floors, and ceiling joists or other structural members where tubing movement is restricted.

4 Construction

4.1 Materials

4.1.1 Component materials

Piping system components shall be constructed entirely of new and unused parts and materials.

Tubing of austenitic stainless steel alloy of the 300 series shall be used.

4.1.2 Examination of components

Components submitted to the testing agency for examination shall be representative production samples.

4.1.3 Composition of materials

The manufacturer shall provide documentation of the composition of the materials used in all components of the piping system. Any materials used that do not comply with a nationally recognized standard shall be certified by a third-party testing agency.

Note: Such documentation is not required for components such as gas pressure regulators, manual gas valves, and quick-disconnect devices that comply with nationally recognized standards.

4.1.4 Suitability of materials

All materials shall be evaluated and found to be suitable for their intended use. Test data based on ASTM or other appropriate test procedures, certifications, or historical data shall be submitted for this purpose. The evidence shall show that the materials have been evaluated, as appropriate, for resistance to moisture, corrosion, the effects of general household chemicals, and the effects of fuel gases, including sulphur compounds therein, and that non-metallic materials exposed to the atmosphere are resistant to the effects of ozone.

4.1.5 Corrosion testing

Tubing and fitting components shall be subjected to the test procedures in ASTM B117 for not less than 96 h without pitting, flaking, cracking, or signs of corrosive attack.

4.1.6 Jacket colour

The colour of the tubing coating, jacket, or covering shall be either yellow or black, and marked in a contrasting colour.

4.1.7 Fire testing of jacket material

For tubing which includes a coating, jacket, or covering, the coating, jacket, or covering shall be tested for flame spread and smoke density as specified in ASTM E84 or CAN/ULC-S102, or equivalent. [See also Clause [4.7](#) I).]

4.2 General

4.2.1 Other assembly requirements

The construction of parts not specifically covered by this Standard shall be in accordance with industry-accepted concepts of safety, substantiality, and durability.

All specifications as to construction set forth herein shall be satisfied by the construction actually prescribed or such other construction as will provide at least equivalent performance, as determined by the listing agency.

4.2.2 Pipe threads

Fittings intended for connection to threaded pipe shall have taper pipe threads conforming to ASME B1.20.1.

4.2.3 Defects

Tubing and fittings shall be clean and free from dents, flaws, or other defects.

4.2.4 Tightness of fittings

Mechanical tube fittings shall be designed to resist loosening from vibration, torquing, and tensile loads. All mechanical tube fittings shall comply with the performance provisions in Clause [5.13](#).

4.3 Gas pressure regulators

Gas pressure regulators supplied by the manufacturer for use with CSST systems shall comply with a nationally recognized standard for pressure regulators.

4.4 Manually operated gas valves

Manually operated gas valves supplied or specified by the manufacturer for use with CSST systems shall comply with a nationally recognized standard.

4.5 Quick-disconnect devices and gas convenience outlets

A quick-disconnect device or gas convenience outlet supplied or specified as part of a gas piping system shall comply with ANSI Z21.41 • CSA 6.9 or CSA/ANSI Z21.90 • CSA 6.24, as applicable.

4.6 Striker plates

4.6.1 Protection from penetration

Striker plates shall be provided by the piping system manufacturer to protect the installed tubing from penetration by nails, screws, etc., in those areas where the tubing will be concealed and will not be free to move to avoid such penetration. (See Clause [5.12](#).)

4.6.2 Design of striker plates

Striker plates shall be designed to protect the tubing at points of penetration through a stud, joist, plate, sill, or other structural member.

4.6.3 Striker plate installation

Striker plates shall include a means for securely mounting/attaching the striker plate to a permanent part of the structure.

4.7 Installation instructions

Complete detailed instructions, including appropriate illustrations, necessary for proper sizing, installation, inspection, and repair of the piping system shall be provided by the manufacturer. These instructions shall be included in a printed design and installation manual to be provided with the piping

system. The manual shall include instructions on all topics and in the exact prescribed format shown in Annex [A](#).

The instructions shall include the following statements (or equivalent) and requirements:

- a) The CSST installer shall be qualified to the manufacturer's training program (see Annex [B](#)).
- b) A warning to the installer that the installation instructions shall be followed as prescribed by the manufacturer.
- c) The installation shall be made in accordance with local codes or, in the absence of local codes, in accordance with one of the following: NFPA 54/ANSI Z223.1, CSA B149.1, the ICC's *International Fuel Gas Code*, IAPMO/ANSI UPC 1, 24 CFR Part 3280, ICC/ANSI 2.0, NFPA 58, or NFPA 501A, as applicable.
- d) Precautions shall be taken by the installer to ensure that any exposed tubing is not damaged or abused during building construction or reconstruction.
- e) The piping system is for use with fuel gases only and is intended for operating pressures not exceeding 5 psi (34.5 kPa) or 25 psi (172.5 kPa). The maximum actual operating pressure, including transients, shall not in any case exceed 6.5 psi (44.8 kPa) for 5 psi (34.5 kPa) rating or 30 psi (207 kPa) for 25 psi (172.5 kPa) rating.
- f) If the piping system installation requires components in addition to those supplied by the piping system manufacturer, the specific components required shall be specified. Only the components provided or specified by the manufacturer or permitted by the local code authority as equivalent are to be used in the installation.
- g) The size and depth of installation clearance holes or notches for routing the tubing through wall studs and joists shall comply with the requirements of the local building code.
- h) Concealed tubing shall be protected from puncture threats, using the shielding devices specified by the manufacturer, at all points of penetration through studs, joists, plates, or similar structures. The extent of protection shall be defined as follows:
 - i) At points of penetration less than 2 in (50.8 mm) from any edge of a stud, joist, plate, etc., a listed striker plate is required to provide protection at the area of support and within 5 in (127 mm) of each side (if appropriate) of the support.
 - ii) At points of penetration 2 to 3 in (50.8 to 76.2 mm) from any edge of a stud, joist, plate, etc., a listed striker plate is required to provide protection throughout the area of support.
 - iii) At points of penetration greater than 3 in (76.2 mm) from any edge of a stud, joist, plate, etc., no protection is required.
 - iv) Tubing routed horizontally through studs shall be protected from puncture threats between the studs using the shielding devices provided.
 - v) CSST greater than 1 in (25.4 mm) inside diameter installed within hollow cavity walls of 2 × 4 construction shall be protected along the entire concealed length in the manner and using the shielding devices specified by the manufacturer.
 - vi) The width of the installed striker plate, at the points of penetration through wall studs, floor joists, plates, sills, etc., shall extend at least 1.5 times the outside diameter of the tubing.
- i) The inspection, testing, and purging of the installation shall be performed using the procedures specified in Part 4 of NFPA 54/ANSI Z223.1, CSA B149.1, the ICC's *International Fuel Gas Code*, and IAPMO/ANSI UPC 1, or in accordance with the requirements of the applicable local codes. The installed gas piping system shall not exhibit any loss of pressure during the field pressure test.
- j) The minimum permissible bend radius for the tubing, and minimum permissible bend radius for fitting/tubing combinations, shall be provided. A warning to avoid sharp bends, stretching, kinking, or twisting of the tubing shall also be provided.
- k) Sizing tables for 0.5 psi (3.45 kPa), 2 psi (13.8 kPa), 5 psi (34.5 kPa), 10 psi (69 kPa) (if applicable), and 25 psi (172.5 kPa) (if applicable) service as a minimum shall be provided to aid the installer or contractor in selecting the proper size tubing for the system.

- l) Flame spread and smoke density ratings for coating materials (as determined by the test procedures listed in Clause [4.1.7](#)) shall be provided. For a flame spread rating greater than 25 for flame spread and/or a smoke density rating greater than 50 for smoke density, the instructions shall include a statement that the installer shall meet local building codes with respect to flame spread and smoke density regulations.
- m) Minimum requirements for bonding of the CSST system (as described in Annex [C](#)) shall be provided, including, but not limited to,
 - i) location of connection;
 - ii) method of connection;
 - iii) size of bonding conductor; and
 - iv) connection to the grounding electrode system.
- n) Information on acceptable regulators, including the size required and any special installation requirements, shall be provided. The regulator(s) specified shall comply with a nationally recognized standard in accordance with Clause [4.3](#).
- o) Guidelines for installing and protecting tubing located outdoors (above ground) shall be provided. The guidelines shall place limits on outdoor installations to those required to make connections to gas meters or outdoor gas appliances which are attached to, mounted on, or located in close proximity to the building structure.
- p) When an excess flow valve is supplied as part of the gas piping system, the CSST manufacturer's design and installation instructions, or instructions supplied with the part by the valve manufacturer, shall include data on sizing and pressure drop across the device as a function of flow (up to the activation flow rate) for each size valve.
- q) For operating pressures above 5 psi (34.5 kPa), engineering guidance for sizing CSST systems shall be provided. The sizing data shall be provided in the form of actual test data, tables, graphs, or formulas for natural gas and propane as determined by the manufacturer.
- r) CSST shall not be directly routed into a metallic gas appliance enclosure utilizing a metallic vent that penetrates a roofline. The CSST connection shall be made outside of the metallic gas appliance enclosure to a section of rigid metallic pipe, stub-out, or termination fittings.
- s) For CSST with an arc-resistant jacket or covering system, the maximum permissible amount of jacket removal shall be stated. For excessive jacket removal or mechanical damage to the jacket, instructions for the repair or replacement shall be provided.
- t) When fabricated tubular manifolds, other than cast or steel pipe, are offered as a component of the piping system, the manufacturer shall include installation instructions describing the requirements and procedures for supporting the manifold when installing the manifold to the piping system and when installing fittings within the manifold ports. Markings shall be provided in accordance with Clause [4.8.8](#).

4.8 Marking

4.8.1 Tubing, manifold, and striker plate markings

Tubing, manifold, and striker plates shall bear the following markings:

- a) manufacturer's name, trademark, or symbol;
- b) part number; and
- c) symbol of the organization making the test for compliance with this Standard.

4.8.2 Permanent markings

Tubing shall bear the following permanent markings:

- a) rated pressure of 5 lb/in² or 25 lb/in² (34.5 kPa or 172.5 kPa);
- b) the EHD (see Clause [5.10.3](#));

- c) the words “FUEL GAS”*;
* The equivalent French wording is “GAZ COMBUSTIBLE”.
- d) the words “Arc-resistant”* or acronym “AR” for CSST design-certified to the requirements in Clause 5.16; and
* The equivalent French wording is “Résistant à l’arc”.
- e) identification with the following marking:
“CSA/ANSI LC 1 • CSA 6.26(:year)”.

4.8.3 Marking intervals

The markings specified in Clauses 4.8.1 and 4.8.2 shall appear on the tubing at intervals of 24 in (610 mm) or less.

4.8.4 Qualified installer marking

Each factory coil of CSST shall bear a marking stating that the CSST must be installed by a manufacturer’s qualified installer.

4.8.5 Fitting marking

Each fitting of a piping system shall bear the following markings:

- a) manufacturer’s name, trademark, or symbol;
- b) fitting size code; and
- c) symbol of the organization making the test for compliance with this Standard.

4.8.6 Marking for packaging

The fitting part number shall be marked on the fitting carton.

4.8.7 Date marking

Each component of the piping system supplied by the manufacturer shall bear a separate date code marking. This marking shall consist of at least four consecutive digits determined as follows:

- a) the first and second digits shall indicate the calendar year in which the component was manufactured (e.g., 22 for 2022); and
- b) the third and fourth digits shall indicate the week in which the component was manufactured (e.g., 03 for the third week of the year). For the purposes of this marking, a week shall begin at 0001 hours on Sunday and end at 2400 hours on Saturday.

A date code shall not be used for more than four consecutive weeks, or for more than two weeks into the next calendar year.

If additional numbers, letters, or symbols follow the four-digit number specified in Items a) and b), they shall be separated from the date code.

4.8.8 Fabricated tubular manifold notice

Where required, each fabricated tubular manifold shall include a label that provides a notice that support is required during installation.

5 Performance

5.1 General

5.1.1 Test temperature

Unless otherwise specified, all performance tests shall be conducted with the piping system components maintained at room temperature, i.e., 77 ± 10 °F (25 ± 5.5 °C).

5.1.2 Test exclusions

The performance tests specified herein shall not be applied to components of a piping system that comply with nationally recognized standards, such as gas pressure regulators, manual gas valves, and quick-disconnect devices, provided that such components are to be installed in accordance with the requirements of nationally recognized standards and the component manufacturer's instructions.

5.1.3 Sample preparation

Unless otherwise specified, tubing having a non-metallic coating or covering shall comply with the tests in Clauses [5.2](#) to [5.16](#) without the coating or covering.

5.2 Leakage

5.2.1 Acceptance criteria

An assembly of tubing and fittings shall not leak externally in excess of 1.22 in³/h (20 cm³/h) of air when tested in accordance with Clause [5.2.2](#).

5.2.2 Method of test

A system of tubing and fittings shall be assembled by joining the components in accordance with the manufacturer's installation instructions. This test assembly shall consist of two of each type of fitting and not less than 6 ft (1.8 m) of tubing. The outlet(s) of the piping system shall be sealed; and the inlet shall be connected to a pneumatic system capable of supplying clean, dry air at the specified test pressure and to a flow-measuring device capable of accurately indicating the allowable leakage rate. The flow-measuring device shall be located between the air supply and the inlet of the piping system. Air shall be admitted to the test assembly and maintained at a pressure of 1.5 times the rated pressure for 1 min, after which the leakage rate shall be measured. The observed leakage rate, corrected to standard conditions of 30 in Hg column pressure and 60 °F (760 mm Hg dry and 15.6 °C), shall not exceed 1.22 in³/h (20 cm³/h).

The leakage criteria permitted in this Clause is for evaluation testing only. Testing for leakage in the field shall be performed in accordance with local codes and Clause [4.7](#) i).

5.3 Pressure capacity

5.3.1 Acceptance criteria

A piping system shall withstand, without bursting or visible loss of fluid, a hydrostatic pressure of 250 psi (1.72 MPa) in accordance with Clause [5.3.2](#).

5.3.2 Method of test

The piping assembly previously tested in Clause [5.2](#) shall be connected to a hydraulic pressure test system, including a pump, a pressure gauge, and heavy-duty pipe fittings, capable of sustaining the

specified test pressure, with care being taken to free all air from the test system and from the piping assembly being tested.

The test pressure shall be increased uniformly and gradually so that a pressure of 250 psi (1.72 MPa) is achieved in approximately 1 min. This pressure shall then be maintained for 1 min, during which time no bursting or visible loss of fluid shall occur.

5.4 Flexibility of tubing

5.4.1 Bending

5.4.1.1 Acceptance criteria

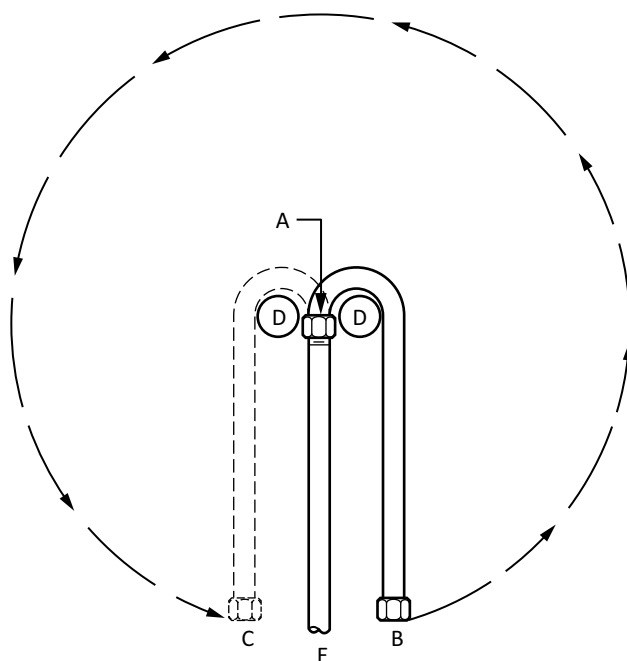
Tubing, fittings, and joints shall withstand 6 bending cycles, in accordance with Clause [5.4.1.2](#), without leakage in excess of that specified in Clause [5.2](#) and without visible damage.

5.4.1.2 Method of test

A length of tubing shall be attached to a fixed pipe to which an air supply system and pressure gauge are connected. The segment length shall be based on a minimum length over internal diameter ratio (L/D) of 24, but shall be at least 2 ft (610 mm) long. Connection to the pipe shall be made using an appropriate fitting in accordance with the instructions provided by the manufacturer. The other end of the tubing shall be closed gas-tight.

Two mandrels, sized to produce the specified minimum bend radius for the fitting/tubing combination (in accordance with the manufacturer's instructions), shall be placed, one at each side of and in contact with the fixed end of the tubing. The mandrels shall be positioned so that their centrelines are opposite the tubing end of the fitting [see Figure [1 a](#)].

Figure 1 a)
Illustration of bending test for fitting/tubing combination
 (See Clause [5.4.1.2.](#))



Legend:

- A = fixed end
- B = position of free end at start of cycle
- C = position of free end at completion of cycle
- D = mandrel
- E = fixed pipe

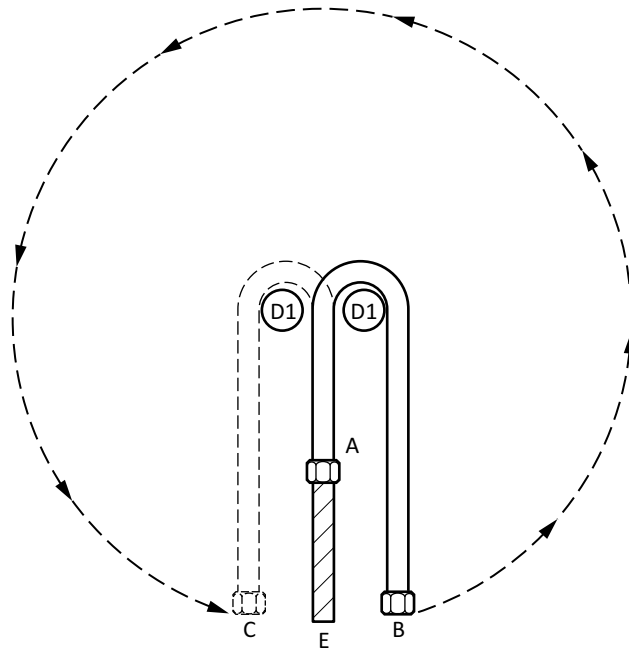
Air shall then be admitted to the tubing until a pressure of 1.5 times the rated pressure is obtained. The tubing shall be bent into the starting position, as illustrated by the solid lines in Figure [1 a\)](#) and the unattached end then bent by hand from the position indicated by "B" along the path shown by the dotted circle to position "C." It shall then be bent back to the starting position. Each bending (from "B" to "C" or from "C" to "B") shall be counted as one cycle. This process shall be repeated until 6 cycles have been completed, the bending motion being applied uniformly at the rate of 5 cycles per minute.

Any leakage as determined by a sudden drop in pressure, as registered on the gauge during the course of the bending operation, shall be considered as non-compliance with this requirement.

Following completion of the 6 cycles, the tubing and fittings shall comply with Clause [5.2.](#)

The test shall be repeated on a new sample of tubing, using the arrangement shown in Figure [1 b\)](#), with two mandrels sized to produce the specified minimum bend radius for the tubing alone (in accordance with the manufacturer's instructions). For this test, the length of tubing between the centreline of the mandrels and point "A" shall be not less than six times the nominal inside diameter of the tubing.

Figure 1 b)
Illustration of bending test for tubing only
 (See Clause [5.4.1.2.](#))



Legend:

- A = fixed end
- B = position of free end at start of cycle
- C = position of free end at completion of cycle
- D = mandrel
- E = fixed pipe

5.4.2 Torsion

5.4.2.1 Acceptance criteria

Tubing, fittings, and joints shall withstand, without leakage in excess of that specified in Clause [5.2](#) and without damage, six applications of 90° (1.57 rad) twists in alternate directions, when tested in accordance with Clause [5.4.2.2.](#)

5.4.2.2 Method of test

A section of tubing whose length is 69 times its actual internal diameter shall be used for this test.

Connections to the tubing shall be made using appropriate fittings and in accordance with the manufacturer's installation instructions.

One end of the tubing shall be securely attached to a rigid frame and the tubing loaded in tension. The load in pounds shall be 50 times the actual internal diameter of the tubing in inches.

For example, for tubing with an actual internal diameter of 0.59 in (15 mm), the length of the test specimen would be 40.7 in ($69 \times 0.59 = 40.7$) [1033 mm ($1752.6 \times 0.59 = 1033$)], and the test load would be 29.5 lb ($50 \times 0.59 = 29.5$) [64.9 kg ($110 \times 0.59 = 64.9$)].

An air pressure of 1.5 times the rated pressure shall be maintained in the system while the tubing is rotated 90° (1.57 rad) at the lower fitting, in a plane perpendicular to the axis of the tubing, and then returned to the original position and rotated 90° (1.57 rad) in the opposite direction. Each 90° (1.57 rad) twist and subsequent return to the original position shall be counted as one cycle, and the twisting motion shall be applied uniformly at a rate of 5 cycles per minute. Following 6 torsion cycles, the assembly shall comply with Clause [5.2](#).

5.4.3 Flow capacity

5.4.3.1 Acceptance criteria

Tubing, fittings, and joints shall withstand, without leakage in excess of that specified in Clause [5.2](#), and without a reduction in flow capacity of more than 50%, a 180° (3.14 rad) twist of the tubing as specified in Clause [5.4.3.2](#).

5.4.3.2 Method of test

New sections of tubing shall be assembled and tested as specified in Clause [5.4.2.1](#), except two samples of tubing shall each be subjected to one twist of 180° (3.14 rad), one sample in a clockwise direction and one sample in a counter-clockwise direction, and held in that position. Each assembly shall then comply with Clause [5.2](#). In addition, the flow capacity shall not have decreased by more than 50% when tested as specified in Clause [5.10](#).

5.5 Axial strength

5.5.1 Acceptance criteria

When tested in accordance with Clause [5.5.2](#), tubing, fittings, and joints shall withstand, without leakage in excess of that specified in Clause [5.2](#) and without becoming detached, a tensile load of 800 lb/in (0.14 kN/mm) of nominal inside tubing diameter.

5.5.2 Method of test

Test segments consisting of tubing and one of each type of fitting shall be assembled in accordance with the manufacturer's installation instructions. The segment length shall be based on a minimum length over internal diameter ratio (L/D) of 6, but shall be at least 6 in (152 mm) long.

One end of the assembly shall be securely attached to a fixed pipe to which an air supply system and pressure gauge are connected. The other end shall be securely attached in a similar manner to a closed pipe connected to a mechanical means capable of applying a constant pulling force of 800 lb/in² (0.14 kN/mm) of nominal inside tubing diameter.

The required tensile load shall be applied to the piping assembly and maintained for the duration of the test. At the end of 5 min, the assembly shall not leak in excess of 1.22 in³/h (20 cm³/h) when tested in accordance with Clause [5.2](#).

5.6 Crushing strength

5.6.1 Acceptance criteria

Tubing and fittings shall withstand, without a decrease in flow capacity of more than 50%, a crushing force of 1000 lb (4.45 kN) when tested in accordance with Clause [5.6.2](#).

5.6.2 Method of test

A test segment of piping shall be assembled as specified in Clause [5.5](#), except each section of tubing shall be not less than 2 ft (610 mm) long.

The assembly shall be placed against a hard, flat surface and a crushing force or weight of 1000 lb (4.45 kN) shall be uniformly applied for 15 min to the tubing, with the force or weight evenly distributed over 6 in (152 mm) of tubing length. The welded seam of the tubing, if applicable, shall be at right angles to the direction of force as viewed along the axis of the tubing. With the weight removed, the flow capacity of the assembly shall then be determined and shall not have decreased by more than 50% when tested as specified in Clause [5.10](#).

A 1000 lb (4.45 kN) crushing test shall also be performed on each type of fitting in the test assembly.

Following the crushing tests, the test assembly shall be tested in accordance with Clause [5.2](#) and shall comply.

5.7 Impact strength

5.7.1 Acceptance criteria

Tubing, fittings, and joints shall withstand an impact of 30 ft•lb (40.7 J) as specified in Clause [5.7.2](#).

5.7.2 Method of test

A piping assembly consisting of each type of fitting and 2 ft (610 mm) long sections of tubing shall be assembled in accordance with the manufacturer's installation instructions. A V-block steel holder shall be used to support the tubing for impacting. The sides of the "V" shall be of sufficient length to support the tubing below the top edges of the "V". The V-block and the fittings to be impacted shall be supported on a hard, flat surface.

An impact of 30 ft•lb (40.7 J) shall be applied to the tubing, using a 10 lb (4.5 kg) weight falling 3 ft (914 mm) through a vertical guide tube. The striking nose of the falling weight shall have a 0.5 in (12.7 mm) spherical radius and shall be made of steel. The arrangement shall be such that the striking nose of the falling weight will impact at the centre of the specimen. A total of five specimens of the tubing shall be impacted, with each specimen being impacted once.

An impact of 30 ft•lb (40.7 J) shall also be applied to each type of fitting in the test assembly, with the testing apparatus arranged so that the centreline of contact between the striking weight and a wrench flat of the fitting is at the longitudinal centre of the flat.

Each test assembly shall then be tested in accordance with Clause [5.2](#) and shall comply.

5.8 Torsion strength of threaded fittings

5.8.1 Acceptance criteria

Fittings joined by threading shall not crack, break, or leak as a result of the application of a tightening torque of 1040 in•lb (117.5 N•m) per inch of nominal tubing inside diameter when tested in accordance with Clause [5.8.2](#).

5.8.2 Method of test

Each type of threaded fitting shall be assembled in accordance with the manufacturer's installation instructions to its mating half or other fitting as intended. One end of the assembly shall be connected leak tight to an air supply system that includes a pressure gauge. The other end of the assembly shall be sealed. Each threaded connection in the assembly, including any pipe thread connections, shall then be tightened by the application of a torque of 1040 in•lb (117.5 N•m) per inch of nominal tubing inside diameter. With wrenches removed, the assembly shall not leak in excess of 1.22 in³/h (20 cm³/h) of air per hour when tested in accordance with Clause [5.2](#).

5.9 Exposure to elevated temperatures (fire hazard resistance)

5.9.1 Acceptance criteria

A piping system shall withstand a temperature of 1000 °F (538 °C) for 1 h, without leaking in excess of 6.0 ft³ (0.17 m³) of air per hour when tested in accordance with Clause [5.9.2](#).

5.9.2 Method of test

A test segment of piping consisting of 1 ft (305 mm) lengths of tubing and one of each type of fitting shall be assembled in accordance with the manufacturer's installation instructions. The outlet of the assembly shall be sealed, and the inlet shall be connected to a pneumatic system capable of supplying clean, dry air at the specified test pressure and to a flow-measuring device capable of accurately indicating the allowable leakage rate. The flow-measuring device shall be located between the air supply and the inlet of the piping assembly. A thermocouple shall be firmly attached to one fitting to monitor its temperature.

The piping assembly shall be placed in a preheated 1000 °F (538 °C) test oven, arranged so that the inlet of the piping assembly extends through an opening in the oven wall and connects to the air supply system.

Air shall be admitted to the system and maintained at a pressure equivalent to the specified maximum actual operating pressure throughout the test, unless a leak is indicated by a drop in pressure. In this event, the test shall be discontinued.

When the temperature of the fitting reaches 990 °F (532 °C), the oven shall be adjusted as necessary so that the fitting temperature is maintained at 990 °F (532 °C) to 1010 °F (543.3 °C) for 1 h. During this period, any leakage indicated by the flowmeter shall not exceed 6.0 ft³ (0.17 m³) per hour when corrected to standard conditions of 30 in Hg column pressure and 60 °F (760 mm Hg dry and 15.5 °C).

5.10 Flow capacity and EHD

5.10.1 Acceptance criteria

The flow capacity for each tubing size shall be equal to or greater than the minimum capacity values for that size as shown in the Tables [1 a\)](#), [1 b\)](#), and [1 c\)](#), and the EHD shall be designated by the testing

organization based on a comparison between the test data and the flow capacity data in Tables [1 a\)](#), [1 b\)](#), and [1 c\)](#).

Table 1 a)
Minimum capacity of CSST in cubic feet per hour for gas pressure of 0.5 psig and a pressure drop of 0.5 in WC (based on 0.60 specific gravity gas)*
 (See Clauses [5.10.1](#), [5.10.3.2](#), and [5.15.2](#).)

EHD flow designation	Tubing length, ft						
	30	50	80	100	150	200	250
13	16	12	9	8	5	4	3
15	23	18	14	13	8	7	5
18	44	34	27	24	19	16	12
19	51	39	31	28	21	19	15
23	75	70	56	50	39	36	32
25	85	75	64	58	45	41	37
30	175	134	105	94	71	66	59
31	203	156	123	110	84	77	68
37	309	233	179	158	126	108	95
39	414	324	259	233	192	168	151
46	674	521	410	367	299	258	230
48	798	620	492	441	361	313	280
60	1423	1107	880	788	646	561	503
62	1584	1229	972	870	711	616	551

* Table includes losses for four 90° bends and two end fittings.

Note: For use only by certification body to validate assigned EHD value.

Table 1 b)
**Minimum capacity of CSST in cubic feet per hour for gas pressure of 2.0 psi and a
pressure drop of 1.5 psi (based on 0.60 specific gravity gas)***
(See Clauses [5.10.1](#), [5.10.3.2](#), and [5.15.2](#).)

EHD flow designation	Tubing length, ft						
	30	50	80	100	150	200	250
13	155	121	96	87	74	63	56
15	200	156	124	111	91	79	71
18	386	303	240	216	177	153	137
19	448	358	283	254	207	180	161
23	702	578	461	415	342	299	268
25	912	711	564	506	415	361	323
30	1637	1299	1020	911	740	639	547
31	1946	1500	1179	1053	856	739	659
37	3007	2345	1866	1674	1374	1194	1071
39	3390	2659	2126	1912	1576	1375	1236
46	6301	4867	3838	3428	2793	2415	2157
48	7058	5487	4352	3898	3192	2770	2481
60	12 411	9663	7676	6881	5641	4900	4392
62	14 296	11 088	8776	7854	6419	5563	4978

* Table includes losses for four 90° bends and two end fittings.

Note: For use only by certification body to validate assigned EHD value.

Table 1 c)
Minimum capacity of CSST in cubic feet per hour for gas pressure of 5.0 psi and a
pressure drop of 3.5 psi (based on 0.60 specific gravity gas)*
 (See Clauses [5.10.1](#), [5.10.3.2](#), and [5.15.2](#).)

EHD flow designation	Tubing length, ft						
	30	50	80	100	150	200	250
13	241	192	153	139	114	99	80
15	322	270	214	191	155	133	119
18	591	460	365	327	268	233	209
19	704	546	432	387	316	274	245
23	1105	866	692	622	513	447	402
25	1385	1078	856	767	629	546	490
30	2583	2007	1577	1407	1143	986	880
31	3005	2315	1820	1624	1320	1140	1017
37	4541	3542	2818	2528	2075	1804	1618
39	5072	3978	3181	2861	2359	2057	1850
46	9669	7469	5889	5261	4286	3706	3311
48	10 717	8331	6608	5920	4847	4206	3768
60	18 797	14 636	11 625	10 422	8544	7421	6652
62	21 792	16 902	13 377	11 972	9785	8480	7589

* Table includes losses for four 90° bends and two end fittings.

Note: For use only by certification body to validate assigned EHD value.

5.10.2 Flow capacity of tubing

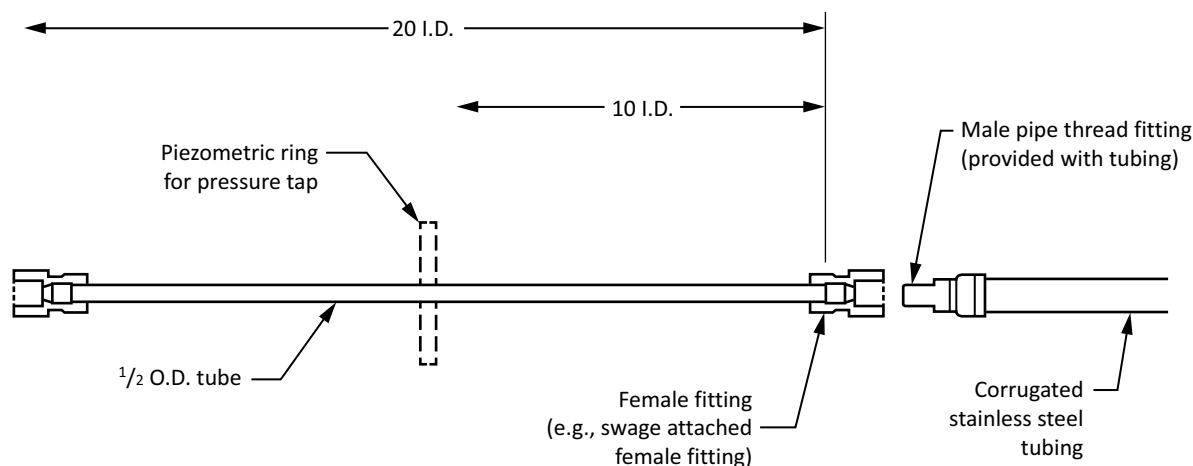
5.10.2.1 Flow capacity

The flow capacity for each tubing size shall be determined as specified in Clause [5.10.2.2](#).

5.10.2.2 Method of test

Standard weight pipe of proper size, reamed to remove burrs caused by cutting, shall be fitted to the inlet and outlet ends of a section of tubing of appropriate length and configuration as shown in Figure 2. The straight run of pipe before and after the tubing shall be of a length not less than 20 pipe diameters (I.D.). Two short lengths of pipe or metal tubing shall be soldered to the pipe, one before the inlet and the other after the outlet connections. These pressure tapings shall be located 10 pipe diameters (I.D.) from the inlet and outlet connections. A drill shall be inserted in the short length of pipe or metal tubing, and a hole drilled through the wall of the large pipe. Care shall be taken to remove any burrs caused thereby. The two pressure tapings shall be connected to a differential pressure gauge, which is directly readable to at least 0.01 in WC (2.5 Pa) pressure.

Figure 2
Piezometer ring detail
 (See Clause 5.10.2.2.)



Either natural gas or air shall be used for the test. If gas is used, it shall be vented or burned as far away from the test equipment as will preclude the heating of such equipment. In the event the flow areas at the inlet and outlet pressure tapings are different, the observed pressure data shall be corrected for differences in velocity head.

For each tubing size, flow rate data shall be measured in accordance with the line pressure, pressure drop, and tubing length conditions specified in Table 2 using the test set-up shown in Figure 3.

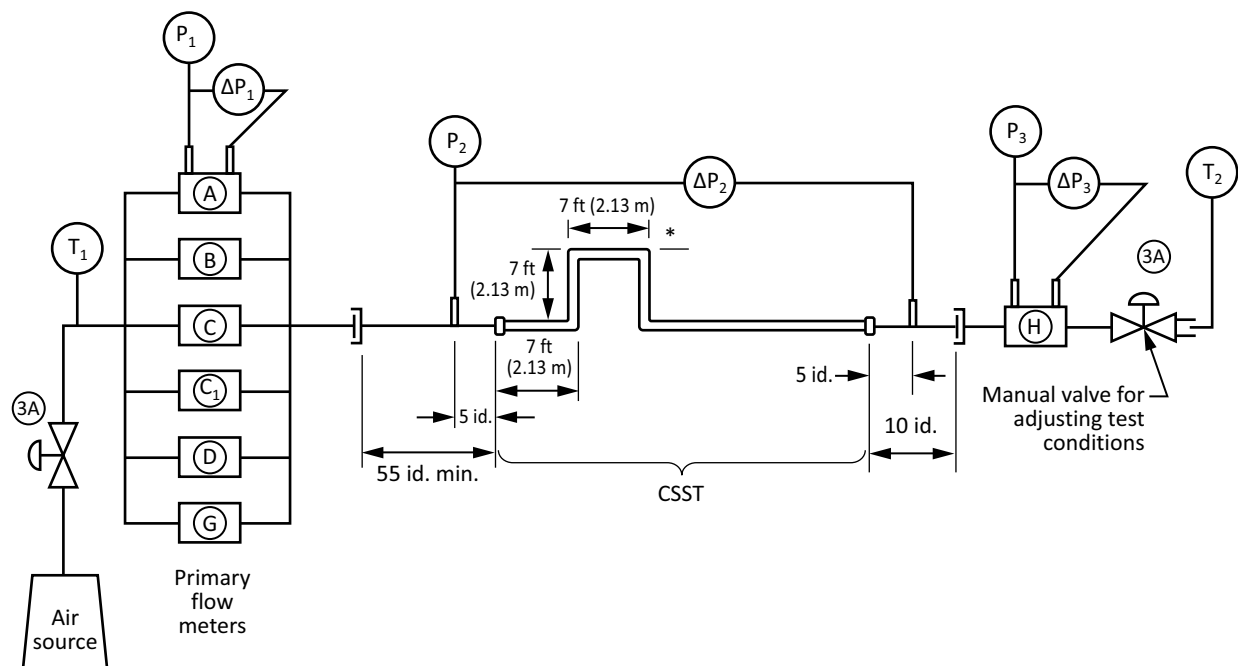
Each series of tests shall begin with a tubing length of 250 ft (76.2 m). If this length is not available, multiple lengths of tubing shall be combined with couplings to achieve this dimension. The inlet pressure shall be adjusted to the value shown in column 1 of Table 2. The flow shall be adjusted until the pressure drop shown in column 2 of Table 2 is achieved. Once steady state conditions have been achieved, the flow rate shall be recorded in the appropriate space in Table 2. The procedure shall be repeated for each combination of line pressure and pressure drop included in columns 1 and 2 of Table 2. The tubing length shall be reduced to the next lower value specified in Table 2, and the test procedure repeated. The room temperature, relative humidity, and barometric pressure shall be measured and recorded in Table 3 after each change of tubing length. Flow capacity data shall be reported in standard cubic feet per hour (SCFH) (m^3/h), corrected for the atmospheric conditions at the time of the test.

The test data shall be reviewed for accuracy and any obvious recording errors. Any data suspected to be irregular shall require a retest of the particular test conditions.

Table 2
Flow capacity data chart
(See Clause [5.10.2.2.](#))

Vendor:								Date:						
Inner diameter:														
Effective hydraulic diameter:														
Column 1				Column 2										
Inlet pressure				Pressure drop				Tubing length ft (m)						
psi	(kPa)	in WC	(kPa)	psi	(kPa)	in WC	(kPa)	30 (9.1)	50 (15.2)	80 (24.4)	100 (30.5)	150 (45.7)	200 (60.9)	250 (76.2)
0.253	(1.74)	7	(1.74)	0.0181	(0.12)	0.50	(0.12)							
0.361	(2.49)	10	(2.49)	0.1085	(0.75)	3	(0.75)							
0.505	(3.48)	14	(3.48)	0.217	(1.49)	6	(1.49)							
2	(13.79)	55.4	(13.79)	1.5	(10.3)	41.49	(10.33)							
5	(34.47)	138.0	(34.48)	3.5	(24.1)	96.81	(24.1)							

Figure 3
Experimental set-up for capacity measurements of test data
 (See Clauses [5.10.2.2](#) and [5.11.2.2](#).)



Legend:

- A = meter A
- 3A = control valve for adjusting test conditions
- B = meter B
- C = meter C
- C₁ = meter C₁
- D = meter D
- G = meter G
- H = meter H
- P₁ = inlet meter pressure
- ΔP₁ = meter reading
- P₂ = inlet tubing (CSST) pressure
- ΔP₂ = pressure drop across tubing (CSST)
- P₃ = meter measuring
- ΔP₃ = pressure drop of control valve
- T₁ = temperature of air before meter
- T₂ = temperature of air at outlet

* Make all bends to minimum recommended bend radius per manufacturer's instructions.

Table 3
Flow capacity data chart — Raw data
(See Clause [5.10.2.2.](#))

Vendor:				Date:					
Inner diameter:									
Effective hydraulic diameter:									
Column 1		Column 2							
Inlet pressure		Pressure drop		Length ft (m)	Meter pressure	Meter temperature	Barometric pressure	Humidity %	Air temperature
psi (kPa)	in WC (kPa)	psi (kPa)	in WC (kPa)						
0.253 (1.74)	7 (1.74)	0.0181 (0.12)	0.50 (0.12)						
0.361 (2.49)	10 (2.49)	0.1085 (0.75)	3 (0.75)						
0.505 (3.48)	14 (3.48)	0.217 (1.49)	6 (1.49)						
2 (13.79)	55.4 (13.79)	1.5 (10.3)	41.49 (10.33)						
5 (34.47)	138.4 (34.47)	3.5 (24.1)	96.81 (24.1)						

5.10.3 Determination of EHD

5.10.3.1 General

The EHD for each tubing size shall be determined and assigned as specified in Clause [5.10.3.2](#).

5.10.3.2 Method of test

The data for each tubing size shall be plotted on a linear scale on the basis of pressure drop (vertical axis) and flow capacity (horizontal axis), as shown in Figure 4. The testing organization shall perform a regression analysis of this data for the purpose of obtaining a best fit curve. The factor of correlation shall be equal to or greater than 0.99. The characteristic equation describing the curve shall be generated in the following power curve form:

$$y = ax^b$$

where

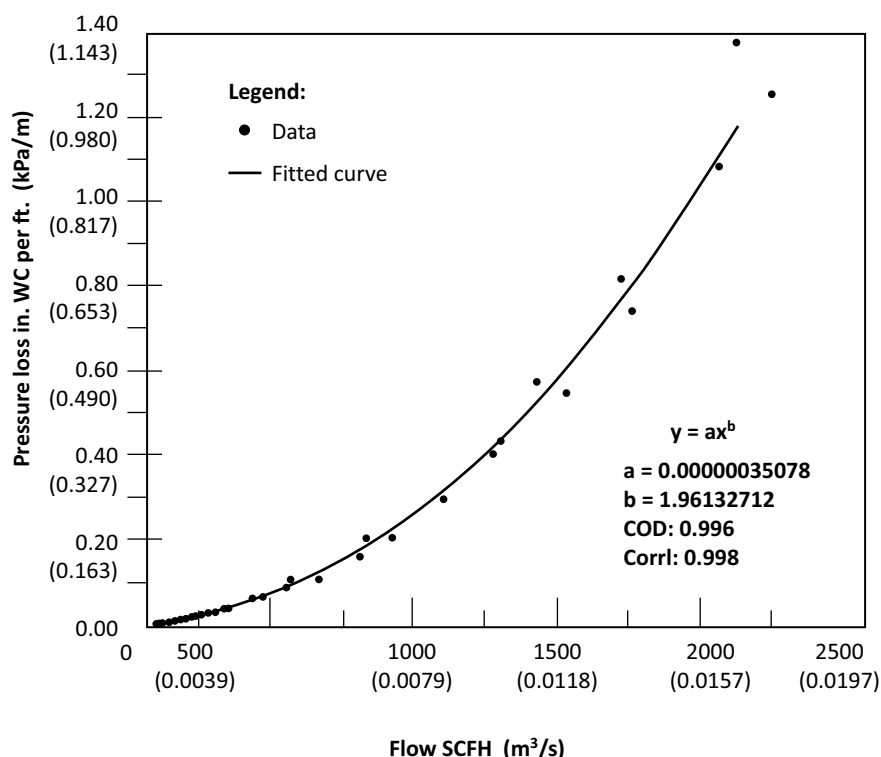
y = pressure drop [in WC/ft (kPa)]

a = calculated coefficient

x = flow capacity [SCFH (cm³/s)]

b = calculated coefficient

Figure 4
Typical plot of test data
(See Clause [5.10.3.2](#).)



Using the characteristic equations, the capacity values shall be calculated. A point-by-point comparison shall be made between the data in Tables [4](#), [5](#), and [6](#) and Tables [1 a](#)), [1 b](#)), and [1 c](#)). The certification body shall designate the appropriate EHD value based on these comparisons. The EHD value assigned to

each tubing size shall be based on the capacity values being equal to or greater than the capacity data (for a particular EHD value) included in the standard sizing Tables [1 a](#)), [1 b](#)), and [1 c](#)). Appropriate engineering judgment shall be exercised to address minor deviations from this requirement.

Table 4
Minimum capacity of CSST in cubic feet per hour for gas pressure of 0.5 psi and a pressure drop of 0.5 in WC (based on 0.60 specific gravity gas)
 (See Clause [5.10.3.2](#).)

Nominal tubing diameter, in	Tubing length, ft (m)						
	30 (9.1)	50 (15.2)	80 (24.4)	100 (30.5)	150 (45.7)	200 (60.9)	250 (76.2)
3/8							
1/2							
3/4							
1							
1-1/4							
1-1/2							
2							

Note: For use only by certification body to validate assigned EHD value.

Table 5
Minimum capacity of CSST in cubic feet per hour for gas pressure of 2.0 psi and a pressure drop of 1.5 psi (based on 0.60 specific gravity gas)
 (See Clause [5.10.3.2](#).)

Nominal tubing diameter, in	Tubing length, ft (m)						
	30 (9.1)	50 (15.2)	80 (24.4)	100 (30.5)	150 (45.7)	200 (60.9)	250 (76.2)
3/8							
1/2							
3/4							
1							
1-1/4							
1-1/2							
2							

Note: For use only by certification body to validate assigned EHD value.

Table 6
Minimum capacity of CSST in cubic feet per hour for gas pressure of 5.0 psi and a
pressure drop of 3.5 psi (based on 0.60 specific gravity gas)
 (See Clause [5.10.3.2.](#))

Nominal tubing diameter, in	Tubing length, ft (m)						
	30 (9.1)	50 (15.2)	80 (24.4)	100 (30.5)	150 (45.7)	200 (60.9)	250 (76.2)
3/8							
1/2							
3/4							
1							
1-1/4							
1-1/2							
2							

Note: For use only by certification body to validate assigned EHD value.

5.11 Pressure drop from bends

5.11.1 General requirements

The pressure drop exhibited by the fittings, manifolds, and 90° (1.57 rad) tubing bends shall not exceed that specified by the manufacturer when the system is delivering natural gas at the specified flow rate(s) in standard cubic feet per hour (SCFH) (cm³/s).

5.11.2 Pressure drop from bends

5.11.2.1 Acceptance criteria

The pressure drop created in a piping system by a 90° (1.57 rad) bend in the tubing, made to the minimum bend radius specified by the manufacturer, shall be determined as specified in Clause [5.11.2.2.](#)

5.11.2.2 Method of test

Standard weight pipe of proper size, reamed to remove burrs caused by cutting, shall be fitted to the inlet and outlet ends of a straight segment of tubing 35 ft (10.7 m) in length. The straight run of pipe before and after the tubing shall be of a length not less than 20 pipe diameters (I.D.). Two short lengths of pipe or metal tubing shall be soldered to the pipe, one before the inlet and the other after the outlet connections. These pressure tapings shall be located 10 pipe diameters (I.D.) from the inlet and outlet connections. A drill shall be inserted in the short length of pipe or metal tubing, and a hole drilled through the wall of the large pipe. Care shall be taken to remove any burrs caused thereby. The two pressure tapings shall be connected to a differential pressure gauge which is directly readable to at least 0.01 in WC (2.5 Pa) pressure.

Either natural gas or air shall be used for the test. If gas is used, it shall be vented or burned as far away from the test equipment as will preclude the heating of such equipment. In the event the flow areas at

the inlet and outlet pressure tapings are different, the observed pressure data shall be corrected for differences in velocity head.

Using the test set-up shown in Figure 3, and with the inlet test pressure held constant at 7 in WC (1.74 kPa) [or greater if required to establish the 150 ft³/h (1180 cm³/s) flow rate], the flow rate shall be adjusted to approximately 150 ft³/h (1180 cm³/s). The measured pressure drop shall be observed and recorded.

The test shall be repeated, except the 35 ft (10.7 m) length of tubing shall first be bent 90° (1.57 rad) in four places as shown in Figure 3. The bending shall conform with the manufacturer's specified minimum bend radius. The centres of the bends shall be spaced 7 ft (2.1 m) apart, starting 7 ft (2.1 m) from the end of the tubing. The pressure drop determined for this configuration, less the total pressure drop determined for the straight configuration, shall be divided by four to determine the average pressure drop for a 90° (1.57 rad) bend. The resultant pressure drop for one 90° (1.57 rad) bend shall not exceed the value specified by the manufacturer for the specified flow rate.

5.11.3 Pressure drop from fittings

5.11.3.1 Acceptance criteria

When tested in accordance with Clause 5.11.3.2, the pressure drop exhibited by each type of fitting (including couplings, straight connectors, and tees) and manifolds shall be determined at the specified flow rate(s) and shall not exceed the manufacturer's declared values.

5.11.3.2 Method of test

The pressure drop test for other fittings (tees) and manifold designs shall be conducted in a similar manner as specified for tubing bends in Clause 5.11.2. Tests shall be conducted with and without additional fittings installed (excluding the termination fittings to connect the tubing to the inlet and outlet pipe) in the 35 ft (10.7 m) length of tubing. The difference in pressure drop between the two tests shall be divided by the number of extra fittings to determine the average pressure drop for one fitting. The resultant pressure drop shall not exceed that value specified by the manufacturer for the specified flow rate.

5.12 Effectiveness of striker plates

5.12.1 Acceptance criteria

Striker plates shall effectively protect the installed tubing from puncture or other damage when impacted by driven nails as simulated in Clause 5.12.2.

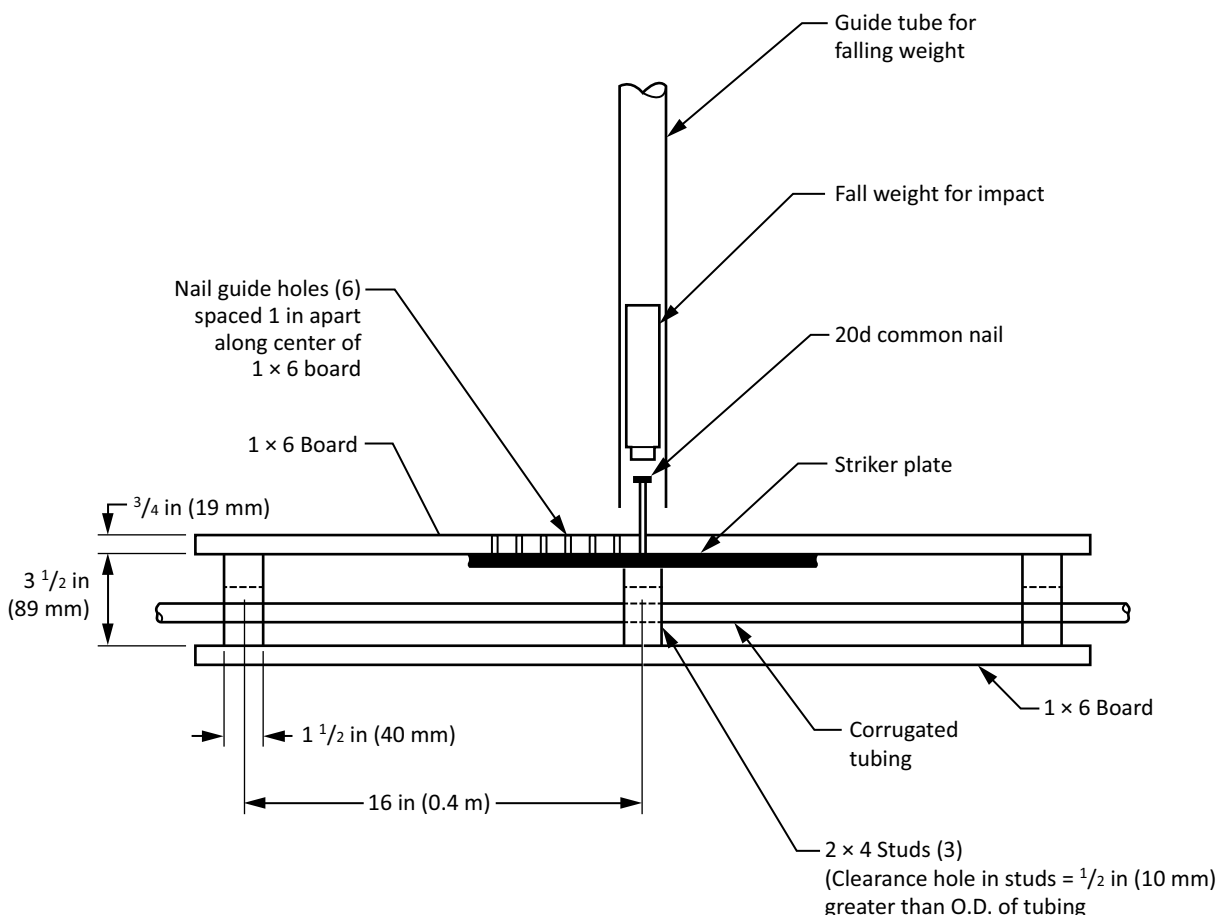
5.12.2 Method of test

This test shall be conducted with any tubing coating or covering in place. All testing shall be conducted using 3/4 in (19 mm) inside diameter tubing.

A straight section of tubing approximately 48 in (1.22 m) long shall be mounted horizontally in a test apparatus that simulates typical wood frame wall construction, as shown in Figure 5. The clearance holes in the studs shall be 1/2 in (12.7 mm) larger in diameter than the outside diameter of the tubing. The striker plate provided by the manufacturer shall be installed at the centre stud in accordance with the manufacturer's installation instructions. A nominal 1 × 6 board approximately 3 ft (914 mm) long shall be attached by nailing to the outer studs so that the board is centred over the tubing and striker plate, as shown in Figure 5. The 1 × 6 board shall be predrilled with nail clearance holes 3/16 in

(4.8 mm) in diameter, at locations corresponding to the centre of the shielded stud, and at 1 in (25.4 mm) intervals from the centre of the stud along the centreline of the length of the striker plate, as shown in Figure 5.

Figure 5
Arrangement for nail puncture test
(See Clause 5.12.2.)



A 20D common steel nail shall be placed in the guide hole of the 1 x 6 board at the centre of the shielded stud. The nail shall be subjected to an impact of 30 ft•lb (40.7 J) using a 10 lb (4.5 kg) weight falling 3 ft (914.4 mm) through a vertical guide tube. The striking nose of the falling weight shall be a flat, smooth surface approximately 1 in (25 mm) in diameter and shall be made of hardened steel. The arrangement shall be such that the nail is impacted squarely and driven perpendicularly to the stud surface and tubing centreline. The impact shall be repeated for a total of 10 strikes. If at any time during this test the nail becomes deformed due to resistance of the striker plate to penetration, the nail shall be replaced as required until the 10 impacts are completed, or until five nails have been deformed, whichever occurs first.

The impact test shall be repeated (as appropriate depending on the size and configuration of the plate) at each of the remaining nailing locations along the length of the striker plate as shown in Figure 5, with 10 impacts (or five deformed nails) being applied at each location. There shall be no nail penetration of the striker plate as a result of these tests. In addition, there shall be no denting or gouging of the tubing.

Following the completion of the test, the coating shall be removed from the tubing specimen and the specimen shall be visually examined for damage. If denting has occurred, and the internal cross-sectional area of the tubing is distorted, then a flow capacity test shall be performed in accordance with Clause 5.10.2. The damage is considered unacceptable if denting results in a reduction in flow capacity of more than 5% of the rated capacity of the tubing specimen, or if metal gouging or wall loss is observed on the tubing surface as a result of the test.

5.13 Mechanical tube fittings — Resistance to loosening

5.13.1 Acceptance criteria

Piping system connections assembled with mechanical tube fittings shall withstand vibration, tension, and torque without loosening and without developing leakage.

The test specified in Clause 5.13.2 shall be applied to each type, material, and nominal size of mechanical tube fitting. New, unused samples of tubing and fittings shall be used for each test.

5.13.2 Method of test

The test shall be conducted as follows:

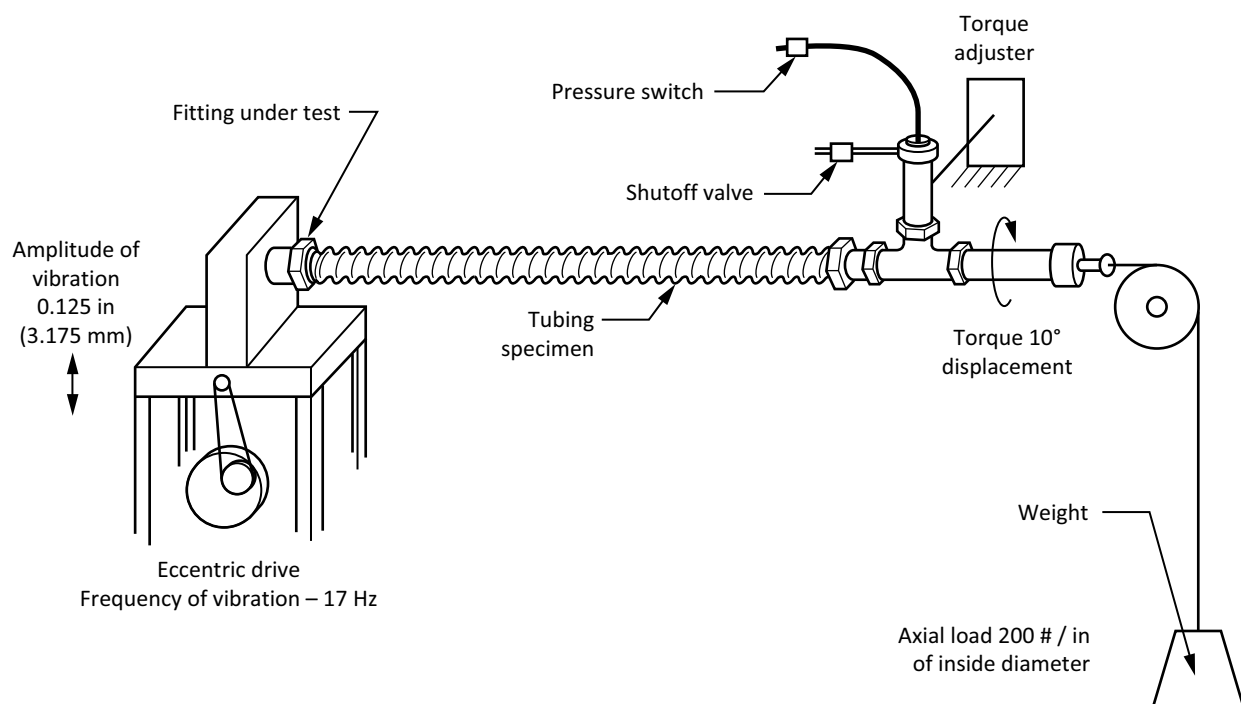
- a) A length of tubing, as shown in Table 7, shall be assembled to fittings at each end in accordance with the manufacturer's instructions. The test shall then proceed as follows:
 - i) The assembly shall be mounted in a test apparatus as shown in Figure 6. One end shall be sealed gas-tight, and the other connected to a pneumatic system capable of supplying clean, dry air at the specified test pressure and to a flow-measuring device capable of accurately indicating the allowable leakage rate. The flow-measuring device shall be located between the air supply and the inlet of the assembly under test.
 - ii) Air shall be admitted to the test assembly and maintained at a pressure of 1.5 times the rated pressure throughout the test. Any leaks in the test apparatus or in the connections to the test apparatus shall be eliminated prior to beginning the test.
 - iii) The tubing assembly shall be loaded in tension with an axial load equal to 200 lb/in (890 N) of nominal internal tubing diameter. The assembly shall also be loaded with a clockwise radial torque, applied at opposite ends of the tubing, as required to produce and maintain a 10° (0.17 rad) displacement between the at-rest positions of the two ends of the tubing assembly. During application of the radial torque, means shall be provided to prevent rotation of pipe threads at the connections of the tubing assembly to the test apparatus.
 - iv) The vibration device shall be adjusted to produce a vibration amplitude of 0.125 in (3.2 mm) at a frequency of 1000 cycles per minute. The amplitude is defined as the maximum displacement of sinusoidal motion from position of rest or one-half the total displacement of the tube fitting.
 - v) The vibration device shall be placed in operation for 55 min, followed by an off period of 5 min. The vibration cycle of 55 min on and 5 min off shall then be repeated continuously for a total test period of 30 consecutive hours. During this test period, the test pressure, tensile load, and torque load shall be applied continuously.
 - vi) Leakage, as indicated by the system flow meter, shall be monitored throughout the 30 h test period. Any leakage in excess of 1.22 in³/h (20 cm³/h), corrected to standard conditions of 30 in Hg column pressure and 60 °F (760 mm Hg dry and 15.6 °C), shall constitute non-compliance with the test.
- b) The test in Item a) shall be repeated except with the tensile load removed.
- c) The test in Item a) shall be repeated except with the radial torque applied in a counter-clockwise direction.

- d) The test in Item a) shall be repeated except with the tensile load removed and the radial torque applied in a counter-clockwise direction.

Table 7
Tubing lengths for vibration, tension, and torque test
 (See Clause [5.13.2.](#))

Tubing ID, in (mm)	Length of test section, in (mm)	Length over diameter ratio
3/8 (9.5)	14 (356)	37
1/2 (12.7)	17 (432)	34
3/4 (19.1)	21 (533)	28
1 (25.4)	26 (660)	26
1-1/4 (31.8)	31 (787)	25
1-1/2 (38.1)	36 (914)	24
1-3/4 (44.5)	40 (1016)	23
2 (50.8)	45 (1143)	23

Figure 6
Test apparatus for the vibration, tension, and torque test
 (See Clause [5.13.2.](#))



5.14 Resistance to outdoor environment

5.14.1 General

The tests in Clause [5.14](#) shall be performed on each nominal size, type, and material of tubing and fitting.

5.14.2 Resistance to extreme temperature cycles

5.14.2.1 Acceptance criteria

Tubing and fitting assemblies shall withstand exposure to extreme temperature cycling without leakage, as specified in Clause [5.14.2.2](#).

5.14.2.2 Method of test

A length of tubing shall be assembled to fittings at each end in accordance with the manufacturer's instructions. The segment length shall be 2 ft (610 mm) long for tubing diameters of 1 in (25.4 mm) or less, and 4 ft (1219 mm) long for tubing diameters greater than 1 in (25.4 mm). One end shall be sealed gas-tight, and the other equipped for connection to a pneumatic system capable of supplying clean, dry air at the specified test pressure and to a flow-measuring device capable of accurately indicating the allowable leakage rate. The flow-measuring device shall be located between the air supply and the inlet of the assembly under test.

A mandrel, with a diameter sized to produce the minimum permissible bend radius for fitting/tubing combinations (in accordance with the manufacturer's instructions), shall be positioned in contact with the tubing at one end so that the centre of the mandrel is aligned with the fitting/tubing interface. The tubing shall then be bent 180° (3.17 rad) around the mandrel as shown in Figure [7 a](#)). The other end of the tubing shall then be bent in the same manner in the opposite direction to form an "S"-shaped assembly as shown in Figure [7 b](#)).

Figure 7 a)
Illustration of bending — One step
(See Clause [5.14.2.2.](#))

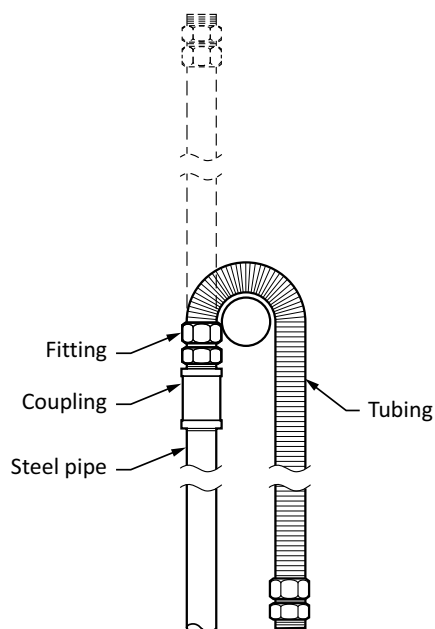
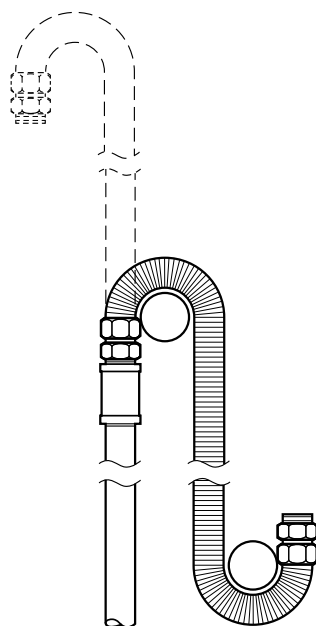


Figure 7 b)
Illustration of bending — Two step
(See Clause [5.14.2.2.](#))



Air shall be admitted to the test assembly and maintained at a pressure of 1.5 times the rated pressure throughout the test. Any leaks in the test system or in the connections to the test assembly shall be eliminated prior to beginning the test. A thermocouple shall be attached to the tube fitting on one end of the test assembly to monitor its temperature.

The assembly shall be mounted in an environmental chamber in the vertical position with the chamber initially at room temperature. The temperature cycling sequence shall be as follows:

- a) The chamber temperature shall be established at 50 ± 2 °F (10 ± 1 °C) within a period of 30 min. When the test assembly reaches a temperature of 50 ± 2 °F (10 ± 1 °C), as indicated by the thermocouple attached to the tube fitting, it shall be maintained at that temperature for 1 h.
- b) The chamber temperature shall be increased to 140 ± 2 °F (60 ± 1 °C) within a period of 30 min. When the test assembly reaches a temperature of 140 ± 2 °F (60 ± 1 °C), as indicated by the thermocouple attached to the tube fitting, it shall be maintained at that temperature for 1 h.
- c) The procedure in Item a) shall be repeated.
- d) The chamber temperature shall be reduced to -40 ± 2 °F (-40 ± 1 °C) within a period of 30 min. When the test assembly reaches a temperature of -40 ± 2 °F (-40 ± 1 °C), as indicated by the thermocouple attached to the tube fitting, it shall be maintained at that temperature for 1 h.

The entire sequence described in Items a) to d) shall be repeated continuously for a total of 25 complete cycles.

Leakage, as indicated by the system flow meter, shall be monitored throughout the test. Any leakage in excess of 1.22 in³/h (20 cm³/h), corrected to standard conditions of 30 in Hg column pressure and 60 °F (760 mm Hg dry and 15.6 °C), shall constitute non-compliance with the test.

Following this test, the test assembly shall be subjected to the stress corrosion cracking tests in Clauses [5.14.4.2](#) a) and b), as applicable.

5.14.3 Resistance to freezing and thawing

5.14.3.1 Acceptance criteria

Tubing and fitting assemblies shall withstand exposure to freezing and thawing without leakage, as specified in Clause [5.14.3.2](#).

5.14.3.2 Method of test

The “S”-shaped tubing assembly and test set-up described in Clause [5.14.2.2](#) shall be used for this test. The assembly shall be mounted in an environmental chamber in the vertical position and pressurized at 1.5 times the rated pressure throughout the test. The freeze-thaw cycle shall be as follows:

- a) With the environmental chamber initially at 50 ± 2 °F (10 ± 1 °C), the entire test assembly shall be sprayed for 1 min with a fine spray of water.
- b) The temperature of the test chamber shall then be reduced to -40 ± 2 °F (-40 ± 1 °C) within a period of 30 min following the spraying. The temperature of the test assembly shall be allowed to cool to -40 ± 2 °F (-40 ± 1 °C), as indicated by the thermocouple attached to the tube fitting. After reaching chamber temperature, the assembly shall be maintained at a temperature of -40 ± 2 °F (-40 ± 1 °C) for 1 h.
- c) The chamber temperature shall be increased to 50 ± 2 °F (10 ± 1 °C) within a period of 30 min. The temperature of the test assembly shall then be allowed to warm to 50 ± 2 °F (10 ± 1 °C), and shall be held at that temperature for 1 h.

The cycle of spraying, freezing, and thawing shall be repeated continuously for a total of 25 cycles. Leakage, as indicated by the system flow meter, shall be monitored throughout the test. Any leakage in excess of 1.22 in³/h (20 cm³/h), corrected to standard conditions of 30 in Hg column pressure and 60 °F (760 mm Hg dry and 15.6 °C), shall constitute non-compliance with the test.

The entire test shall be repeated with the tubing assembly mounted with the bends in a horizontal plane.

5.14.4 Resistance to corrosion

5.14.4.1 Acceptance criteria

Tubing and fitting assemblies shall withstand exposure to a corrosive environment without leakage.

The test assembly previously subjected to the procedure in Clause [5.14.2](#) shall be used for the test specified in Clause [5.14.4.2](#).

5.14.4.2 Method of test

The fittings of the test assembly used in Clause [5.14.2](#) shall be retightened to the manufacturer's specifications. The test assembly shall be pressurized at 1.5 times the rated pressure throughout the test. The pressure source shall be isolated with a shutoff valve, and a pressure gauge shall be used to monitor any sudden pressure drop in the test assembly.

- a) For 300 series stainless steel materials:
 - i) The test assembly shall be immersed in a solution by weight of 20% sodium chloride, 1% sodium nitrate, and 79% distilled water. The temperature of the solution shall be raised to 212 °F (100 °C) at atmospheric pressure and maintained at that temperature for 14 h. All boil-off vapour shall be condensed and returned to the test solution.
 - ii) After 14 h in the boiling test solution, the test assembly shall be removed from the solution and cooled to room temperature. The assembly shall then be reconfigured to a reverse "S" by first straightening the tubing and then bending each end of the tubing in the opposite direction, using the procedure described in Clause [5.14.2](#).
 - iii) Following the test, any leakage in excess of 1.22 in³/h (20 cm³/h), corrected to standard conditions of 30 in Hg column pressure and 60 °F (760 mm Hg dry and 15.6 °C), shall constitute non-compliance with the test.
- b) For copper alloy materials:
 - i) If copper alloy fittings are provided with protective coatings, this test shall be conducted with the coating in place.
 - ii) The test assembly shall be suspended in a sealed plastic vessel containing 8.5 oz (250 mL) of full-strength ammonia (28%) and 8.5 oz (250 mL) of water. Care shall be taken to prevent the test assembly from coming in contact with the liquid solution at any time.
 - iii) After 18 h, the test assembly shall be removed from the container and reconfigured to a reverse "S" by first straightening the tubing and then bending each end of the tubing in the opposite direction, using the procedure described in Clause [5.14.2](#).
 - iv) Following the test, any leakage in excess of 1.22 in³/h (20 cm³/h), corrected to standard conditions of 30 in Hg column pressure and 60 °F (760 mm Hg dry and 15.6 °C), shall constitute non-compliance with the test.

5.15 Electrical properties

5.15.1 Resistance and conductivity

An assembly of tubing and fittings shall not have an electrical resistance that exceeds the values given in Table [8](#).

Table 8
Maximum electrical resistance
 (See Clause [5.15.1](#).)

Nominal size of pipe, in	Resistance, Ω /ft
3/8	0.180
1/2	0.150
3/4	0.120
1.0	0.120
1-1/4	0.090
1-1/2 and 2.0	0.090

5.15.2 Method of test

A length of tubing (with sheathing) shall be laid out (essentially straight) on a flat, horizontal, noncombustible electrically non-conductive surface. The segment length shall be 6 ft (1830 mm) long. Connection to the tubing shall be made using appropriate end fittings, installed in accordance with the manufacturer's instructions. A single, low-resistance copper conductor (no smaller than 6 AWG) of sufficient length shall be secured (by an appropriate means as determined by the certification body) to each end fitting and to an alternating-current supply with a no-load voltage not to exceed 12 V.

A current of 25 A shall be passed between the end fittings of the test assembly by starting with an impressed potential of 0.0 V ac and gradually increasing the voltage until a 25 A current is attained. The test specimen shall not be moved or otherwise manipulated during any part of this test. The voltage drop between the end fittings shall be measured. The resistance shall be calculated from the measured voltage drop and the current in the following equation and compared against the acceptance value listed in Tables [1 a](#)), [1 b](#)), [1 c](#)):

$$R = E/I$$

where

R = ohms

E = volts

I = amperes

5.16 Arc-resistant jacket or covering system (optional)

5.16.1 Jacket requirements

5.16.1.1 Compliance with this Standard

Unless otherwise specified, all testing shall be performed in accordance with the requirements of this Standard.

5.16.1.2 Compliance with installation codes

Tubing that has an arc-resistant jacket or covering system (henceforth referred to as the "jacket") as a manufacturer's alternative to the direct-bonding method (as described in Annex [C](#)) shall comply with

the tests in Clause [5.16](#), and with Clause [4.7](#), including compliance with local or model codes such as CSA C22.1 or NFPA 70, where applicable.

5.16.1.3 Sample preparation

Unless otherwise specified, the jacket shall comply with the tests in Clause [5.16](#), with the tubing, jacket, and fitting(s) assembled as a final installation and in accordance with the manufacturer's instructions.

5.16.1.4 Sample size

Testing shall be performed on samples of tubing sizes 1/2 in and 1 in or as specified herein. Each test sample shall be at least 3 ft (914.4 mm) in length or as specified in Clauses [5.16.2](#) to [5.16.4](#).

5.16.2 Resistance of jacket material to extreme environment(s)

5.16.2.1 Resistance of jacket assembly to extreme temperature cycles

Test specimens of the jacket material shall be prepared (based on the fabrication technique used to apply the jacket) and subjected to exposure to low temperature in accordance and compliance with ASTM D746. The manufacturer shall present a test report from an accredited testing laboratory, including a statement on the minimum embrittlement temperature.

5.16.2.2 Resistance of jacket assembly to corrosion

The tubing, jacket, and fitting assembly (assembled in accordance with the manufacturer's instructions) shall be subjected to the standard practice described in ASTM B117 for not less than 96 h without pitting, flaking, cracking, or signs of corrosive attack in accordance with Clause [4.1.5](#) of this Standard, and then shall be subsequently subjected to and pass the test in Clause [5.16.3](#) of this Standard. Two samples of each size shall be prepared and subjected to the prescribed test. Products that do not incorporate a metallic layer or component as part of the jacket are not required to be tested to this Clause. The manufacturer shall present a test report from an accredited testing laboratory and deliver the test samples to the laboratory performing the electrical tests.

5.16.3 Electrical tests — Robustness against arcing (indirect lightning)

5.16.3.1 General

Tubing, jacket, and fitting assemblies shall withstand electrical arcing, in accordance with Clause [5.16.3.2](#), without perforation of the tubing and without leakage in excess of that specified in Clause [5.2](#). The test samples subjected to the salt spray exposure (if applicable) described in Clause [5.16.2.2](#) shall be used in the testing. Other samples shall be used if no samples are required to be subjected to the salt spray exposure described in Clause [5.16.2.2](#). Testing shall be performed by an accredited lightning testing laboratory.

5.16.3.2 Method of test

5.16.3.2.1 Electrical waveform

For arc robustness determination, an electrical waveform (current vs time) shall be used. The waveform is defined by its rise-time to peak current and fall-time to 50% of peak amplitude. The selected waveform is 10 μ s $\times$ 1000 μ s.

5.16.3.2.2 Generator calibration

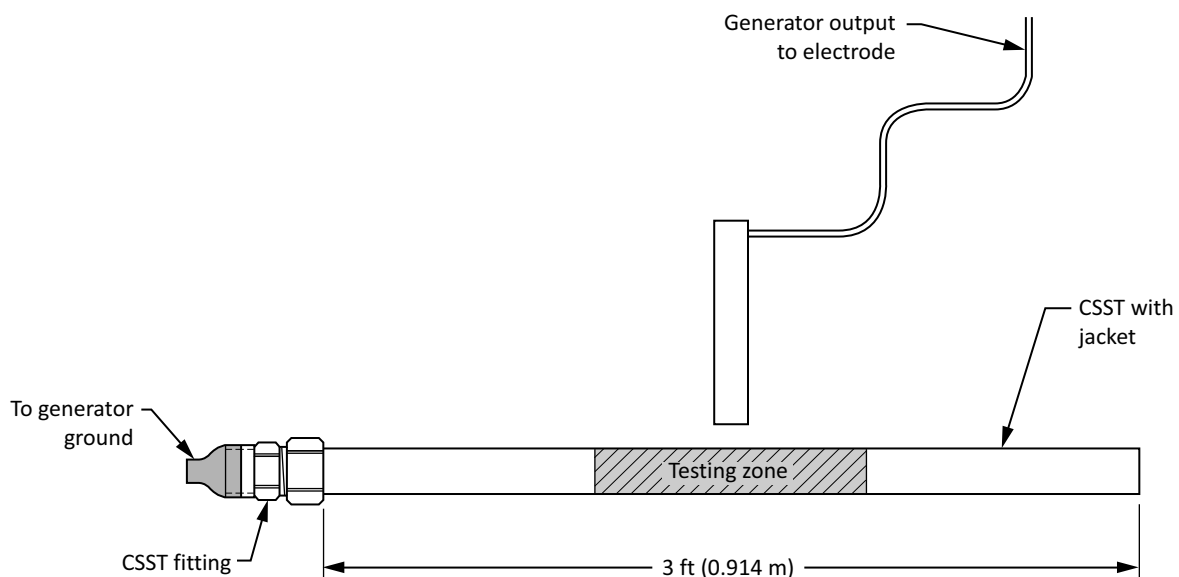
An electrical test generator and appropriate measurement equipment shall be assembled to generate the waveform under consideration and to record generator output. (See Figure 8.) The generator assembly shall use an electrode of 1/4 in (6.4 mm) diameter attached to the output of the generator (i.e., live terminal). The test sample shall be grounded to the generator with a minimum 6 AWG copper conductor or equivalently sized braided strap. The test sample shall be supported in such a manner to prevent arcing or conduction from the test sample to adjacent equipment.

A length of copper pipe shall be installed as the test sample, with a 1/8 in (3.2 mm) gap between the copper pipe and the generator electrode. The generator shall be charged to the desired level, and discharged through the electrode to the copper pipe (via an arc) and then to ground. Arc attachment to the tubing shall be verified. Arc attachment to nearby equipment or to end connections on the copper pipe invalidates this test run. The recorded generator output shall be verified against the desired waveform. Adjustments to the generator assembly shall be made and retested until the generator output matches the desired waveform output.

5.16.3.2.3 Method of test

The test sample shall be installed with the electrode placed over a straight portion of the sample, with a 1/8 in (3.2 mm) gap between the exterior of the jacket and the generator electrode. (See Figure 8.) The generator shall be charged to the desired level and discharged through the electrode to the test assembly (via arc) and then to ground. Arc attachment to the tubing shall be verified. Arc attachment to nearby equipment or to end connections on the tubing invalidates this test run.

Figure 8
Experimental set-up for arc resistance test
(See Clauses [5.16.3.2.2](#) and [5.16.3.2.3](#).)



The recorded generator output shall be verified against the desired waveform. Peak current and coulombs transferred (the integral of the waveform) shall be recorded. The jacket shall be removed from the site of the arc termination, allowing a visual inspection of the exposed tubing. The tubing shall show no puncture of the stainless steel tubing.

The test assembly shall withstand a minimum of 4.5 coulombs without leakage, as defined by Clause [5.2](#).

5.16.4 Resistance to installation damage

5.16.4.1 Acceptance criteria

Tubing and jacket assemblies shall withstand damage from friction/wear in a simulated drag-zone installation without excessive damage to the jacket in accordance with Clause [5.16.4.2](#).

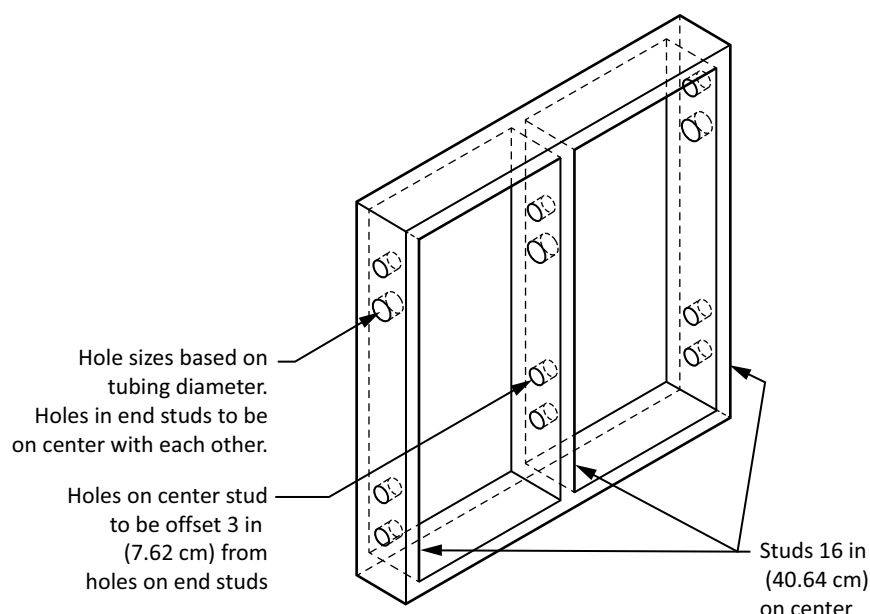
Upon completion of the pull test, the tubing and jacket shall be inspected for physical damage. Any tearing or ripping of the jacket exposing the underlying stainless steel tubing indicates non-compliance with this requirement.

5.16.4.2 Method of test

Three individual simulated joists/studs [spaced 16 in (406 mm) on-centre] made from solid, dimensional lumber (2 × 6 in) shall be mounted in a framework, as shown in Figure [9](#). The drag-zone apparatus can be table or floor mounted. A routing hole shall be drilled through each joist in a 3 in (76.2 mm) offset pattern as shown in Figure [9](#). Each routing hole shall be 1/2 in (12.7 mm) greater in diameter than the outside diameter of the jacketed tubing specimen. The routing holes shall be drilled with a standard hole-cutter/drill bit, and no post-drilling cleaning or dressing of the holes shall be permitted. No metal fasteners shall be installed within the drag zone. The test apparatus shall be solidly secured to the table/floor to prevent movement or shaking of the assembly during the pull tests.

Two test specimens of each size up to and including 1 in (25.4 mm), consisting of a 15 ft (3.8 m) length of tubing, shall be prepared. Each test specimen shall be snaked through the three offset holes before starting the test. The length of tubing shall then be pulled, by hand, through the drag zone until the end of the test specimen exits the last member. The rate of pull should be approximately 1 to 2 ft (304.8 to 609.6 mm) per second.

Figure 9
Set-up for tubing wear test
 (See Clause [5.16.4.2.](#))



6 Manufacturing and production tests

6.1 General

The manufacturer shall submit to the certification body a plan that

- is mutually acceptable to the manufacturer and the certification body; and
- describes the programs and test procedures specified in Clauses [6.3](#) and [6.4](#) and the records to be kept by the manufacturer.

6.2 Qualified purchased components

The manufacturer shall use a program to qualify raw materials, parts, assemblies, and purchased components.

6.3 Test requirements

The manufacturer shall test each coil of corrugated tubing for leakage.

6.4 Schedule of tests

The manufacturer shall use a program that includes a mutually acceptable schedule(s) to conduct

- bending tests (Clause [5.4.1](#));
- torsion tests (Clause [5.4.2](#)); and
- strength tests (Clauses [5.5](#), [5.6](#), [5.7](#), and [5.8](#)).

The manufacturer's test method(s) specified in Clause [6.4](#) shall be capable of relating back to the test(s) specified in this Standard. The section numbers of tests in this Standard are provided in parenthesis for ease of reference.

7 Items unique to Canada

7.1

All markings and instructions required by this Standard shall be provided in a form that is easily understood in both the English and French languages. See Table 9.

Table 9
English and French markings
(See Clause 7.1.)

Clause number	English	French
4.8.2 c)	FUEL GAS	GAZ COMBUSTIBLE
4.8.2 d)	Arc-resistant	Résistant à l'arc

7.2

Units of measurement required on printed instructions and markings shall include the SI units as a minimum.

Annex A (normative)

Minimum design and installation manual requirements

Note: This Annex is a mandatory part of this Standard.

A.1

A printed design and installation manual shall be provided with the piping system (see Clause [4.7](#)). The manual shall include instructions on all topics and in the exact prescribed format shown below:

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6.0 Pressure/leakage testing

Requirements for low pressure systems

Requirements for elevated pressure systems

7.0 Sizing tables (natural gas and propane)

7.1 CSST

7 in/0.5 in WC drop

8 to 10 in/3 in WC drop

11 to 13 in/6 in WC drop

11 in/0.5 in WC drop (propane only)

2 psi/1 psi drop

5 psi/3.5 psi drop

10 psi/7 psi drop (if applicable)

25 psi/10 psi drop (if applicable)

7.2 Steel pipe

7 in/0.5 in WC drop

8.0 Definitions

Annex B (normative)

Installer training

Note: *This Annex is a mandatory part of this Standard.*

B.1

The manufacturer shall establish and maintain an installer training program and a database of installers who have completed the manufacturer's training requirements. An identification card shall be supplied to each installer who has completed the manufacturer's training requirements, and a record of the installer's contact data shall be maintained in the database. The identification card shall include the following information:

- a) CSST manufacturer's name and telephone number;
- b) CSST product(s)/brand(s) covered by the training;
- c) unique serial number for the identification card (recorded in the database);
- d) installer's name and the date of training or card issuance (recorded in the database); and
- e) statement that the named installer has completed the manufacturer's training program to install [insert manufacturer's name or brand] CSST.

Annex C (informative)

Instructions for the direct (electrical) bonding of CSST piping systems

Note: *This informative Annex has been written in mandatory language to facilitate adoption by anyone wishing to do so.*

C.1 Purpose

Bonding the CSST (corrugated stainless steel tubing) piping system to the grounding electrode system of the structure in which it is installed will lower the unintended voltage build-up on the CSST caused by indirect sources such as power surges and lightning strikes. Bonding will help achieve an electrically neutral condition between the CSST and other similarly bonded metallic systems such as the water piping, structural steel, and coax cable, and will help reduce the probability and severity of arcing between these conductive systems when energized by a lightning strike near the structure.

This Annex describes the requirements for the bonding of corrugated stainless steel tubing for fuel gas piping systems, which shall be performed by a qualified person recognized by the local AHJ as capable of performing such work. Direct bonding is required as part of the installation of all new and retrofit fuel gas piping systems that contain CSST not complying with the requirements of Clause [5.16](#). This includes all fuel gas systems, whether or not the connected equipment is electrically powered. The requirements are applicable to typical single-family and multi-family dwellings and certain commercial structures.

C.2 Conditions

C.2.1

CSST not complying with Clause [5.16](#) installed inside or attached to the exterior of a building or structure shall be electrically continuous and directly bonded to the electrical ground system of the structure. The gas piping is considered to be directly bonded when installed in accordance with the following conditions:

- a) A bonding conductor is permanently and directly connected to the electrical service grounding system. This can be achieved through a connection to the electrical service equipment enclosure, the grounded conductor at the electrical service, the grounding electrode conductor, or to the one or more grounding electrodes used. All grounding electrodes used shall be bonded to the electrical service grounding electrode system or, where provided, the lightning protection grounding system.
- b) A single-bond connection is made to the building gas piping system on a rigid piping component or CSST fitting located downstream of the utility gas meter or second-stage LP regulator. A daisy-chain configuration of the bonding conductor is permitted for multi-meter installations. The bonding clamp attachment point may be at any location within the gas piping system. The shortest practical bonding conductor length will improve the effectiveness of the direct-bond, but the bonding conductor length shall not exceed 75 ft (22.86 m). A bonding connection shall not be made to the underground natural gas utility service line or the underground supply line from an LP storage tank.
- c) The bonding conductor shall not be smaller than a 6 AWG copper wire or equivalent. The bonding conductor shall be installed and protected in accordance with NFPA 70 and CSA C22.1. Where a larger bonding conductor is required, it shall be sized in accordance with NFPA 70, Table 250.66.
- d) The bonding conductor shall be attached in an approved manner in accordance with NFPA 70 and CSA C22.1, and the point of attachment for the bonding conductor shall be accessible.

- e) The bonding/grounding clamp used shall be listed to UL 467/CSA C22.2 No. 41-07 or other acceptable national standards.
- f) The bonding clamp shall be attached at one point within the piping system to a segment of rigid pipe or a pipe component such as a nipple, fitting, manifold, or CSST fitting. (See Figures [C.1](#) and [C.2](#).) The corrugated stainless steel tubing portion of the gas piping system shall not be used as the point of attachment of the bonding clamp at any location along its length. The bonding clamp shall be attached such that metal-to-metal contact shall be achieved with the pipe component.

Figure C.1
Bonding attachment to pipe
 (See Clause [C.2](#).)

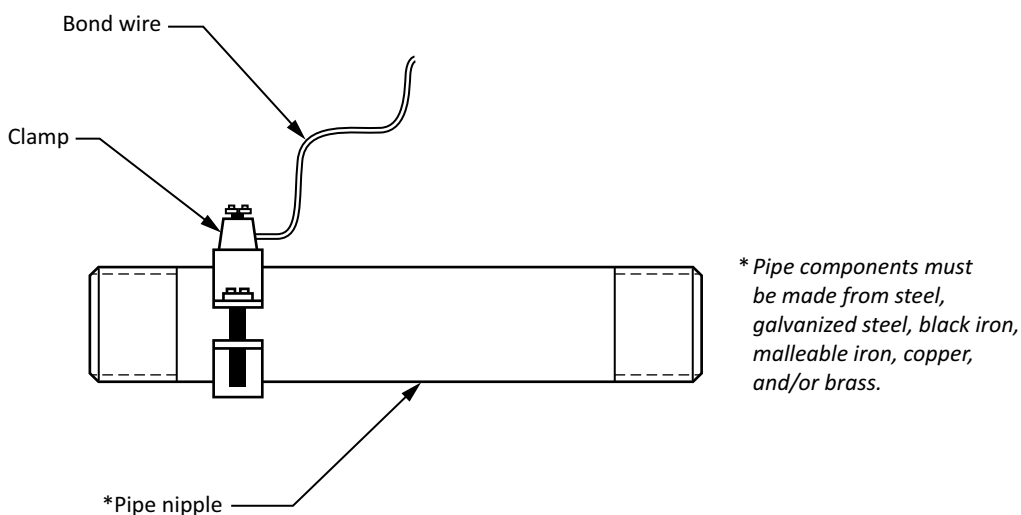
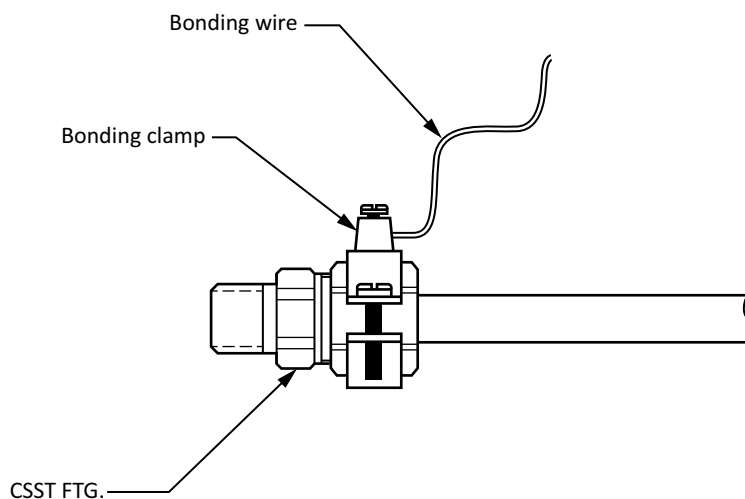


Figure C.2
Bonding clamp on CSST fitting
 (See Clause [C.2](#).)



Note: CSST should not be directly supported on or by other electrically conductive systems including copper water pipe, electric power cables, air conditioning and heating ducts, communications cables, and structural steel beams.

C.2.2

CSST complying with Clause [5.16](#) installed inside or attached to the exterior of a building or structure shall be electrically continuous. CSST compliant with Clause [5.16](#) shall be considered to be bonded when the piping system is connected to an appliance within the system that is connected to the grounding conductor of the circuit supplying that appliance. If the fuel gas piping system containing CSST complying with Clause [5.16](#) is not connected to an appliance utilizing an equipment grounding conductor, or the piping system contains any amount of CSST that does not comply with Clause [5.16](#), the fuel gas piping system shall be directly bonded in accordance with Clause [C.2](#).

Annex D (informative)

Conversion factors

Note: This Annex is not a mandatory part of this Standard.

D.1 Temperature scales and conversions

The unit of temperature in the International System of Units (SI) is the kelvin (K), but it is generally accepted practice to express temperature differences in terms of degrees Celsius (°C) because the degree intervals are identical. The term “centigrade” was abandoned in 1948 by the General Conference on Weights and Measures but in fact is still in common use. The accepted abbreviation for centigrade is also °C and, for all practical purposes, the degree intervals of centigrade, Celsius, and kelvin are identical.

Many temperature measurements are still made in terms of degrees Fahrenheit (°F). Although a formal definition of the Fahrenheit scale does not exist, it is based on

- a) the freezing (ice) point of water = 32 °F;
- b) the boiling point of water under standard pressure conditions = 212 °F;
- c) the formula for absolute temperature, $5/9 (°F - 32) = °C$; and
- d) the formula for “temperature rise,” $5/9 °F = °C$.

See Table [D.1](#).

Table D.1
Temperature scales and conversions
(See Clause [D.1](#).)

°C	°F	°C	°F	°C	°F
-40	-40.0	25	77.0	70	158.0
-20	-4.0	30	86.0	80	176.0
0	32.0	35	95.0	90	194.0
10	50.0	40	104.0	100	212.0
15	59.0	50	122.0	110	230.0
20	68.0	60	140.0	120	248.0

D.2 Multiples and submultiples of SI units

Table [D.2](#) lists the primary multiples and submultiples of SI units.

Table D.2
Multiples and submultiples of SI units
 (See Clause [D.2.](#))

Factor by which the unit is multiplied			Prefix	Symbol
1 000 000 000 000	=	10^{12}	tera	T
1 000 000 000	=	10^9	giga	G
1 000 000	=	10^6	mega	M
1000	=	10^3	kilo	k
100	=	10^2	hecto	h
10	=	10^1	deca	da
0.1	=	10^{-1}	deci	d
0.01	=	10^{-2}	centi	c
0.001	=	10^{-3}	milli	m
0.000 001	=	10^{-6}	micro	μ
0.000 000 001	=	10^{-9}	nano	n
0.000 000 000 001	=	10^{-12}	pico	p

